

From Memory to Testable Belief: A Survey of LLM Agents

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Abstract

Language-model agents now act over many steps, and the failures reported from such settings are failures of state rather than of reasoning. The agent loses track of what it has learned, keeps acting on contradicted evidence, or revises what it believes without changing what it does. What an agent must hold between decisions is not a memory of what happened but a testable belief, a hypothesis that later evidence can test and revise, with three parts, a state that says what is believed and how firmly, a revision that carries it across an action, and a test that scores it against the next observation. This survey takes the belief state as its object and sorts the literature by which of the three parts a system has and where each comes from, whether written by the model, fixed by the designer, learned, or absent. Memory and uncertainty research gave the agent a belief state, and revision mechanisms and post-training taught it to revise. The test, the part that lets the agent's own observations correct what it believes, is supplied from outside wherever it exists. We follow the three parts through construction, revision, and testing, align belief-level evaluation by construct, and identify the gap the alignment exposes. The first systems to score a written belief against the agent's own observations do so with a binary belief, in isolation, and without a framework that connects representation, testability, revision, credit, and belief-action consistency.

AI disclosure. The literature search, screening, and coding pipeline released with this paper is scripted and uses large-language-model agents to read and code each sampled paper under a fixed codebook (Appendix A); large-language-model assistance (Claude, Anthropic) was also used in drafting and revising the manuscript. The authors defined the framework, the inclusion criteria, and the codebook, checked the codings that support

Table 1 | Which part of the belief each neighbouring survey supplies.

	Survey	Organising axis	S	R	T	E
Memory	Zhang et al. (2024b)	design, evaluation, use	■	■	■	■
	Hu et al. (2025d)	factual / experiential / working	■	■	■	■
	Luo et al. (2026c)	storage → reflection → experience	■	■	■	■
Reward	Wu (2025)	reward model × stage	■	■	■	■
	Zheng et al. (2025a)	data → PRM → use	■	■	■	■
	Zhang et al. (2025d)	capability × application	■	■	■	■
	Zhang (2026)	granularity × method	■	■	■	■
Uncertainty	Xia et al. (2025)	four estimation avenues	■	■	■	■
	Lin et al. (2026c)	what / how / where	■	■	■	■
Evaluation	Yehudai et al. (2026)	capability × domain	■	■	■	■
	Gao et al. (2025)	what / when / how to evolve	■	■	■	■
	Chen et al. (2026b)	lifecycle × horizon carrier	■	■	■	■
	Xu et al. (2026d)	forecasting architecture	■	■	■	■
	Dong et al. (2025b)	financial function	■	■	■	■
Belief	Cai et al. (2026)	control-loop components	■	■	■	■
	Tang et al. (2026c)	decision pipeline × policy	■	■	■	■
This survey		belief state · revision · test	■	■	■	■

Evaluation includes horizon, self-evolution, forecasting and finance surveys, and Belief those that formalise a Bayes-filter belief for one setting. S, the belief state and its persistence; R, revision, how the state is revised; T, a test that reaches the belief as a learning signal; E, belief-level evaluation. Solid, central; tint, present; grey, absent.

the claims in the text, and are responsible for the content.

1 Introduction

Language-model agents now act over horizons of hours or months, and the failures reported from these settings are failures of state, not of reasoning. A model that solves a problem in one pass loses much of its accuracy when the same information arrives over several turns (Laban et al., 2025), an agent given only observations wins a fraction of the episodes won by one that reads the simulator's latent state (Alaswad et al., 2026), and in markets stronger backbones do not reliably produce stronger trading agents (Xie et al., 2024; Li et al., 2024a). What these failures share is that the task required the agent to hold something between decisions, a belief about what is true, how certain it is, and what would change it, and the agent never formed it, never tested it, or revised it without acting on the revision.



Fig. 1 | Taxonomy of belief-state work. Work is split by the three parts of a testable belief, each branch by where the part comes from. The highlighted leaf holds two systems, both without uncertainty.

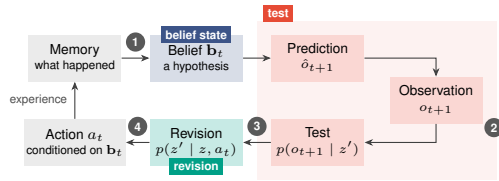


Fig. 2 | From memory to testable belief. A memory records what happened, whereas a belief is a hypothesis that prediction, observation, and test expose to evidence. Tints mark the three parts. Any evidence-based check can serve as the test, and the formulas give one realisation, the Bayes filter of Eq. 1. Numbers mark the failure modes of Section 5 (1 isolation, 2 stay, 3 update, 4 act).

That something is a testable belief, and this is the shift the survey argues for. A memory records what has happened, whereas a belief is a hypothesis about what is true of the world, which a prediction exposes to evidence, a test supports or falsifies, and a revision updates (Fig. 2). The test can take many forms (Section 3). This review follows three parts, a *belief state* that says what is believed and how firmly, a *revision* that carries it across an action, and a *test* that scores it against the next observation, with the Bayes filter as their principled formalisation (Åström, 1965; Kaelbling et al., 1998; Thrun et al., 2005; Friston, 2010). The line runs through memory research, not around it. A store that attaches a confidence and an update rule to its claims already has the first two parts (Zhang et al., 2024b; Luo et al., 2026c), and what makes it

a belief is the third, a test that connects the hypothesis to evidence that can support or falsify it. Memory and uncertainty research gave the agent a state with a confidence (Park et al., 2023; Shinn et al., 2023; Lin et al., 2022; Kadavath et al., 2022), and post-training taught it to revise, rewarding the response, the step, or the action (Ouyang et al., 2022; Lightman et al., 2024; Zhou et al., 2024b; Zhang, 2026). The test is still supplied from outside, by a fixed observation model, a simulator, a verifier, or a teacher.

Work on the belief itself exists, from dialogue state tracking (Lee et al., 2021) to agents that write beliefs about tasks, opponents, and markets (Kim et al., 2024; Cross et al., 2024; Yu et al., 2024) and benchmarks that score the belief rather than its outcome (Xu et al., 2026a; Deb and Krishnan, 2026; Chao et al., 2026), but it rarely cites itself (Fig. 6) and the nearest surveys organise by what is stored, stop at the action, or formalise a Bayes-filter belief for one setting at inference time only (Section 2, Table 1). This survey takes the belief state as its object and sorts the literature by which of its three parts a system has and where each comes from, whether written by the model, fixed by the designer, learned, or absent (Fig. 1), and follows the belief around the cycle of Fig. 2. Section 4 asks where the state lives and what form it takes, Section 5 how it is carried and revised and where revision fails, and Sections 6–8 who scores the belief against evidence, whether the score be-

comes a learning signal and *which object receives it*, and how evaluation measures each part. We cover 2020 to September 2026 by a scripted protocol (Appendix A), and Section 10 states the open problems.

2 Related Work

Table 1 places this survey among sixteen neighbouring ones by which part of the belief each supplies. Memory surveys (Zhang et al., 2024b; Hu et al., 2025d; Luo et al., 2026c) supply the belief state and its persistence, the newest with a probability or revision operator on each item, but they do not ask what evidence would retire an item, which is the test. Uncertainty surveys (Xia et al., 2025; Lin et al., 2026c) supply a confidence on the state, but attached to an output and neither carried forward nor revised. Reward and credit surveys (Wu, 2025; Zheng et al., 2025a; Zhang et al., 2025d; Zhang, 2026) supply the revision and the learning signal, following credit down to tokens, steps, and memory operations, and stop at the action. World-model and long-horizon surveys (Li et al., 2026d; Chen et al., 2026b) supply a learned revision and test whose state the agent cannot read. Evaluation, self-evolution, forecasting, and finance surveys (Yehudai et al., 2026; Gao et al., 2025; Xu et al., 2026d; Dong et al., 2025b) score the outcome and treat the belief as one component among many. Two recent surveys formalise the belief itself as a Bayes filter, for complex problem solving (Cai et al., 2026) and for proactive service agents (Tang et al., 2026c), but update it at inference time in one setting, so the test never reaches it as a learning signal. Each strand holds one part or one setting, and none follows the three parts across agents into learning and evaluation, which is the question Section 3 turns into a definition.

3 Problem Formulation

A belief $\mathbf{b}_t$ is a hypothesis about the environment that the agent carries between decisions. It yields a prediction, evidence o_{t+1} returns, a *test relation* supports or falsifies the hypothesis, and a revision produces $\mathbf{b}_{t+1}$ for the next action (Fig. 2). The test may be a symbolic verifier, a consistency check, or a comparison with an external record. Partially observed decision-making gives this cycle its most principled realisation (Åström, 1965; Kaelbling et al., 1998). With hidden state z_t and history H_t , the belief is $\mathbf{b}_t = p(z_t \mid H_t)$, and a

Bayes filter carries it forward as

$$\mathbf{b}_{t+1}(z') \propto p(o_{t+1} \mid z') \sum_z p(z' \mid z, a_t) \mathbf{b}_t(z) \quad (1)$$

(Thrun et al., 2005). Its three factors are the Bayesian form of the three parts the review follows. The factor $\mathbf{b}_t$ is the *belief state*, what is believed and how firmly, the transition $p(z' \mid z, a_t)$ is the *revision* that carries it across the action, and the likelihood $p(o_{t+1} \mid z')$ is the *test*. A verifier or consistency check is a non-probabilistic test of the same kind and is coded as one. Agentic reinforcement learning casts the language-model agent as such a POMDP, with the context as history and any explicit state in a working memory the model writes (Zhang et al., 2025d; Sumers et al., 2023; Yao et al., 2023b), and we write $\mathbf{b}_t$ for whatever explicit state the span carries before the action a_t .

We call such an object a *belief state* only if (i) it is a claim about the environment carried between decisions, (ii) it carries, or admits, uncertainty, and (iii) it is *testable*, committing to a prediction that a returned observation can support or falsify. Conditions (i) and (ii) are properties of the belief state and fix the operation CONSTRUCT. Condition (iii) requires a test and fixes TEST, which makes the state a hypothesis rather than a record. REVISE is the revision, which belief-revision theory constrains to change the claim without discarding the rest (Alchourrón et al., 1985; Friston, 2010). We therefore place every system by which of the three parts it has and where each comes from, whether written by the model, fixed by the designer, learned from data, or absent. A trace *memory* fails (i) and (ii). A store that attaches a probability and an update rule to each item meets them but scores nothing against what returns, so it fails (iii) (Luo et al., 2026c). A *world model* is itself a revisable hypothesis, a learned revision and test whose state the agent cannot read, so it meets (iii) but seldom (i) (Li et al., 2026d).

4 The Belief State: Where It Lives and Who Writes It

The belief state is placed not by method family but by two attributes, its *locus* (in the context, in a store outside the model, in an external estimator, or in learned dynamics), which also says who writes it, and its *form* (a trace, a claim, or a distribution). The classes in Table 2 are locus–form pairs, so a system can occupy more than one, and the last column records which of conditions (i)–(iii) each

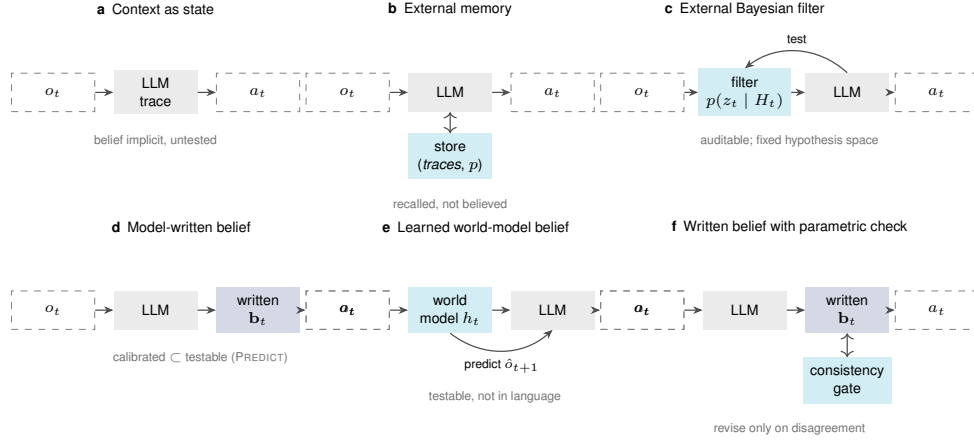


Fig. 3 | Who maintains the belief. Blue-grey marks a belief the model writes and can reason about, light blue an external estimator. Only c, d (when testable), e and f commit to a prediction that later evidence can falsify.

Table 2 | Classes of the belief state and where each part comes from.

Locus	Form	State	Uncert.	Rev.	Test	Belief	Example
Context	trace as state	M		M	—		ReAct
	written claim, calibrated	M		M	D (target)		Agent-BRACE, BOND
	written claim, testable	M		M	D (oracle)		CBM, EvoSCM
Store	trace store	M		D	—		MemGPT, A-MEM
	claims with probabilities	M		D	D (update rule)		Nous, BeliefMem
Estimator	Bayesian filter, fixed space	D		D	D (closed form)		Belief-State Engine
Learned	latent world model	L		L	L (prediction error)		Dreamer, BB-WM
	written claim + parametric check	M		M	L (check)		Hybrid-WM

Locus: in the context, in a store outside the model, in an external estimator, or in learned dynamics. M, written by the model; D, fixed by the designer; L, learned; —, absent (parts of Section 3). Uncert., the state carries uncertainty; Belief, meets conditions (i)–(iii) of Section 3. Solid, yes; tint, partly; grey, no.

class meets. Fig. 3 sketches the six designs, and Table 15 in the appendix places every system.

4.1 Context and Memory

The most common design has no representation at all. The reasoning trace is the state (Wei et al., 2022b; Yao et al., 2023b), the belief implicit and its uncertainty unstated.

Memory-augmented agents moved the state into a store retrieved into context (Packer et al., 2023; Park et al., 2023; Zhong et al., 2024; Xu et al., 2025d; Luo et al., 2026c; Jiang et al., 2026a). Two limits matter for belief. The stored unit is a trace rather than a claim with a probability, and an update is an operation on the store rather than on a hypothesis, so answer accuracy hides failures in the update and forget operations themselves (Hao et al., 2026). Structured stores narrow the first limit with schemas, graphs, bitemporal transitions, or AGM revision semantics (Luo et al., 2026d; Xu, 2026; Park, 2026), and probabilistic stores narrow the second by attaching a number to each claim (Liao et al., 2026a; Song and Xin, 2026; Yang et al., 2026c; Zhang et al., 2026j).

4.2 External Bayesian Filters

A different line takes the belief out of the model. The model maps evidence to a likelihood over a fixed set of latent regimes, hypotheses, or user states, and an explicit filter maintains the posterior, whose entropy and drift can then be reported as model risk (Chattopadhyay and Halder, 2026; Dixon, 2026b; Wu et al., 2026d; Soru, 2026; Akbar et al., 2026; Azran et al., 2026; Ansell and Toney-Wails, 2026). These beliefs are validated against a closed-form oracle and auditable by construction, at the price of a hypothesis space fixed in advance.

4.3 Model-Written Beliefs

The third line keeps the belief in context but gives it structure the model writes and reads. Dialogue state tracking wrote a slot-value belief (Lee et al., 2021), verbalised confidence attached a probability to an answer (Lin et al., 2022; Kadavath et al., 2022), agents wrote beliefs about tasks, opponents, and markets (Kim et al., 2024; Christakopoulou and Mourad, 2024; Cross et al., 2024; Yu et al., 2024), MEM1 learned to overwrite a fixed written state each turn (Zhou et al., 2025d), ReAct reflected on a goal state before each action (Kim et al., 2025), and CoEx kept a neurosymbolic belief that co-evolves with exploration (Kim and won Hwang, 2025). The newest systems add uncertainty and verification to the written object, from certainty tags and Bayesian-teacher posteriors to symbolic verification and populations of causal models (Singh et al., 2026b; Cui and Lin, 2026; Xu et al., 2026a; Zhao et al., 2026b).

Condition (iii) separates them. A written belief

Table 3 | Four failure modes of belief revision.

Failure mode	Definition	Part at fault	Measured by	Headline result
Failed stay	belief moves without evidence	test absent: update unweighted	BeliefShift, MEDLEY-BENCH	consensus sensitivity from near 0 to large across 35 models
Failed update	belief survives contradicting evidence	carried forward, not updated	DeltaLogic, DToM-Track, STALE	66.7% on a conclusion, 46.7% on whether an edit changes it
Failed isolation	irrelevant input enters the belief	belief state contaminated	BeliefTrack, MemOps	belief-state reward cuts failures by 71%
Failed act	belief revised, action unchanged	policy not conditioned on b_t	STOCKTAKE, Werewolf audit	34–43% of correctly diagnosed weeks end in the wrong action

For each mode, the part at fault, the benchmarks that isolate it, and their headline results (Xu et al., 2026a; Chao et al., 2026; Myakala et al., 2026; Abtahi et al., 2026; Dhanda, 2026; Nguyen et al., 2026; Hao et al., 2026; Deb and Krishnan, 2026; Gao et al., 2026b).

is testable only if it names a future observation with a horizon, so that the returned observation yields a signed error. Most written beliefs are calibrated but not testable, stating confidence without naming what would falsify it, whereas Hypothetical Minds, CBM, EvoSCM, and ReBel (Tang et al., 2026b) name it.

4.4 Learned World Models

The fourth line learns the dynamics and exposes the resulting belief to the policy, through recurrent latent states (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2023), the language model as its own world model for tree search (Hao et al., 2023; Zhao et al., 2023b; Zhou et al., 2023a), LLM-proposed POMDPs refined to a belief-based likelihood (Six et al., 2026), and a world-model belief the policy can query (Kumar et al., 2026; Li et al., 2026d; Zuo et al., 2026). World-model beliefs are testable by construction but, unless written back, cannot be reasoned about in language. Hybrids use the learned model as a check on a written belief, and a parametric consistency gate cuts the hallucinated-state rate from 0.176 to 0.035 (Song and Cai, 2026a).

Insight 1. Structure is not testability. Memory design has made the belief steadily more structured without making it testable, and the representations that are testable, filters and world models, cannot be reasoned about in language. The belief the field needs is written and testable at once, and the first system to test one against the environment does so with a binary belief.

5 Revision and Its Failures

Bayesian filtering and active inference agree on how a belief should change. It predicts, evidence arrives, and the mismatch drives the update (Friston, 2010; Friston et al., 2017), routed by factor

graphs to the factors that produced the prediction (Kschischang et al., 2001; Loeliger, 2004). In Equation 1 the transition carries the belief forward and the likelihood decides how far the observation moves it. The language-model literature imported the vocabulary but rarely the routing, appending feedback to the context or folding it into self-critique loops (Huang et al., 2022b; Madaan et al., 2023; Kim et al., 2023a).

5.1 Four Failure Modes

Contextual Belief Management names three ways a belief can fail on new input, a *failed stay* in which the belief moves without evidence (an update no test weighs), a *failed update* in which it survives evidence against it (carried forward without an update), and a *failed isolation* in which irrelevant input contaminates it (Xu et al., 2026a). In consolidated memory a quarter or more of the agent’s own speculations resurface as user-stated facts (Verbovoy, 2026). Failed update is also the base model’s default. Assistants revise correct answers under pushback (Sharma et al., 2023; Atwell et al., 2025), drift as context accumulates (Geng et al., 2025), and update in context like a filter that forgets exponentially (Zhang et al., 2025e), while memory systems write the misconception in (Bensal et al., 2026). Independent studies document a fourth, the *failed act*, in which the belief is revised but the policy is not conditioned on it, so that a belief-level oracle finds the diagnosis right while the action stays wrong (Deb and Krishnan, 2026; Gao et al., 2026b). Table 3 and Fig. 2 place them.

5.2 Mechanisms of Revision

The mechanisms that respond to failed update fall into three families by what triggers them. The first is triggered by *contradiction* among stored claims and resolves it with operator algebras, first-class update and forget operations, or supersession contracts (Wang, 2026b; Hao et al., 2026; Xu, 2026; Qin and Lu, 2026). The second is triggered *before commitment*, by self-audit, a parametric veto, semantic entropy, or resampling (Yuan et al., 2026a; Song and Cai, 2026a; Yan et al., 2026b; Wang et al., 2026b). These see no future observation and so cannot catch a belief that is consistent and wrong. The third is triggered by *surprise*. Reward-prediction error routes writes to a fast or a slow path (Song and Xin, 2026), and prediction–observation mismatches become semantic rules (Zhang et al., 2026g). FinCon revises its invest-

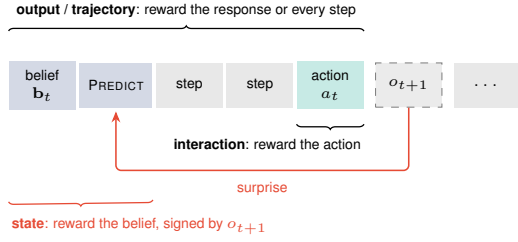


Fig. 4 | The credited object on one span of a trajectory. State credit reaches the written belief, signed by the observation returned after the action.

Table 4 | Learning signals by credited object and source of the test.

Level	Credited object	Test from	Representative
Output	finished response	human preference	RLHF, DPO
Trajectory	sampled steps	verifier, PRM	PRMs, GRPO, LATS
Interaction	actions, tool calls	task reward	ArCher, RAGEN, HIPER
	memory operations	task reward, verifier	MEM1, Memory-R1, VeriMem
State	written belief	symbolic verifier	CBM
	claim + certainty	calibration target	Agent-BRACE
	posterior over claims	Bayesian teacher	BOND, DEL-ToM
	state prediction	simulator state	VAGEN, Dyna-Think, SPA
	belief over answers	known answer	T3, IGPO, InfoPO, CORE, GAVEL
	written belief	own next observation	ReBel
	written belief	own past observations	ABBEL

The object that receives the gradient and where the test that signs it comes from (Section 6). Bold, the test comes from the agent’s own observations. Table 13 in the appendix adds granularity and evidence level.

ment beliefs after each episode and sends each revision only to the agents that need it, routing the error to an agent though not yet to a claim (Yu et al., 2024).

Insight 2. Revision is triggered, not routed. Contradiction, self-audit, and surprise decide *when* a belief changes, but none decides *which claim* the error belongs to. Until the error is routed to the claim that made the prediction, agents will keep repairing memories while beliefs stay wrong.

6 The Test as a Learning Signal

A learning signal is a test that has become a gradient. Fewer than three in five included papers give the belief any such signal, and for the rest it is built and revised at inference time. Where existing surveys organise post-training by reward source and density (Wu, 2025; Zhang, 2026; Zheng et al., 2025a), we separate two axes that are usually run together (Fig. 4, Table 4), the *credited object*, the quantity that receives the gradient, and the *source of the test*, whatever signs it (humans, a verifier, the task reward, a teacher, a simulator, or the agent’s own later observation). A method is at the *state level* if the credited object is a claim about the environment, whatever the source, and the source then decides whether the sign exists outside a simulator.

6.1 Output and Trajectory

Supervised fine-tuning and preference optimisation credit the completed response (Ouyang et al., 2022; Wei et al., 2022a; Rafailov et al., 2023; Wang et al., 2022; Huang et al., 2022a). Trajectory-level credit began by imitating and searching over agent trajectories (Chen et al., 2023a; Zhou et al., 2023a; Putta et al., 2024; Fu et al., 2025), then credited sampled steps through PPO, GRPO, and process reward models (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024; Zheng et al., 2025a).

6.2 Interaction

Agentic reinforcement learning credits actions over several turns (Zhou et al., 2024b; Wang et al., 2025f; Zhou et al., 2025c; Feng et al., 2025b; Xiong et al., 2025a), and the open question is how to make that credit denser without making it wrong, by hierarchical advantages, counterfactual estimators, potential-based shaping, or verifier-structured credit (Peng et al., 2026a; Wang et al., 2026a; Diao et al., 2026; Wang et al., 2026j; Ren et al., 2026a; Liao et al., 2026b; Liu et al., 2026b; Li et al., 2026b). The nearest this level comes to the state is a group that credits the write to a memory, from a rewritten case bank to an RL-trained overwrite of a fixed written state (Zhou et al., 2025a,d; Yu et al., 2025b; Yan et al., 2025; Sun et al., 2026b; Yan et al., 2026a). These credit the *operation*, so a well-executed update to a wrong claim is still rewarded.

6.3 State

Until 2026 every such method took its test from outside the environment, from a symbolic verifier (Xu et al., 2026a), a calibration target (Singh et al., 2026b), a Bayesian teacher (Cui and Lin, 2026; Wu et al., 2025d), the simulator’s state (Wang et al., 2025d; Yu et al., 2025c; Chen et al., 2025e; Wu et al., 2025a), or a known answer (Zou et al., 2025; Wang et al., 2025a; Kong et al., 2026; Zhang et al., 2026j; Wang et al., 2026e). Each needs the answer or the latent state at training time (Appendix D). Three systems now take it from the agent’s own observations. ReBel rewards each predicate of a written belief that the next observation supports, with the gradient reaching the belief (Tang et al., 2026b), ABBEL grades a natural-language belief by how well the prior observation can be reconstructed from it (Lidayan et al.,

Table 5 | Evidence-acquisition decisions by what they are conditioned on.

Acquisition conditioned on	Systems	Belief object
task-level POMDP	CGDP (Kausik et al., 2026), MedExAgent (Gao et al., 2026a)	searcher, patient state
uncertainty of an explicit belief	BayesEvolve (Wu et al., 2026d), EvoSCM (Zhao et al., 2026b), Align-While-Search (Bae et al., 2025), CA-BED (Arnould et al., 2026)	hypothesis, causal, configuration posteriors
belief over own resources	Bayesian-Agent (Wu et al., 2026b), REDEREF (Hosseini et al., 2026), CMI (Srivastava, 2026b), REFLEX (Wu and Lim, 2026)	skill, delegate, memory reliability
information gain as reward	PICA (Liu et al., 2026b), TIPS (Xie et al., 2026)	implicit
written testable belief	<i>none in corpus</i>	—

2025), and PaW adds a next-observation loss on the policy (Lu et al., 2026a). None carries uncertainty, and the dense form is fragile. The same reward through group-normalised advantages collapses every run, and as an auxiliary loss it is matched by shuffled targets (Wang, 2026a).

Insight 3. Credit has reached the belief, not yet its uncertainty. The first systems now score a written belief against the agent’s own observations, but the belief they score is binary, the dense form of that reward collapses under group normalisation, and none connects the score to revision or action. The open combination is an environment-signed test on a calibrated state.

7 The Test as a Choice of Evidence

A testable belief tells the agent what evidence would change it, which turns retrieval, tool use, and routing into decisions (Table 5). Expected information gain needs a probabilistic test, so the systems that choose evidence by belief are those that carry one. Search and diagnosis have been framed as POMDPs whose actions include what to observe next (Kausik et al., 2026; Gao et al., 2026a), and two finance simulators show why it matters, as failing agents spend 62% of their observation budget re-reading internal state (Han et al., 2026; Zhang et al., 2026c).

Agentic retrieval let the model decide when to retrieve (Gao et al., 2024), and recent systems condition that decision on measured uncertainty about the task (Wu et al., 2026d; Zhao et al., 2026b; Xu and Fekri, 2026; Yan et al., 2026b; Bae et al., 2025; Arnould et al., 2026; Wang et al., 2026e), make information gain the training signal itself (Liu et al., 2026b; Xie et al., 2026), or keep a belief over the agent’s own skills, delegates, or memories (Wu et al., 2026b; Hosseini et al., 2026; Srivastava, 2026b; Wu and Lim, 2026).

	calibration	accuracy	revision	belief-act.	outcome
Forecast-Dojo (Ye et al., 2026)	●	○	●	○	●
Agent-BRACE (Singh et al., 2026b)	●	○	○	○	○
Belief-State Engine (Chattopadhyay and Halder, 2026)	●	○	○	●	○
Belief at Risk (Dixon, 2026b)	●	○	○	○	○
FinBench (Ghosh and Devarakonda, 2026)	○	○	○	○	●
BeliefTrack (Xu et al., 2026a)	○	●	○	○	○
DeltaLogic (Dhanda, 2026)	○	○	○	○	○
DTolM-Track (Nguyen et al., 2026)	○	●	○	○	○
BeliefShift (Myakala et al., 2026)	○	○	○	○	○
STALE (Chao et al., 2026)	○	○	○	○	○
MemOps (Hao et al., 2026)	○	○	○	○	○
STOCKTAKE (Deb and Krishnan, 2026)	○	○	○	○	○
Werewolf audit (Gao et al., 2026b)	○	○	○	○	○
HERCULEAN (Peng et al., 2026b)	○	○	○	○	○
FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024a)	○	○	○	○	○
BayesBench (Samanta et al., 2026)	○	○	○	○	○
Martingale Score (He et al., 2025)	○	○	○	○	○
Bayesian coherence (Imran et al., 2025)	○	○	○	○	○
Theory of Space (Zhang et al., 2026f)	○	○	○	○	○
Halumem (Chen et al., 2025b)	○	○	○	○	○
Memora (Uddin et al., 2026)	○	○	○	○	○
Embodied Agent Interface (Li et al., 2024b)	○	○	○	○	○

Fig. 5 | Belief-level constructs reported by each evaluation paper. Only the Bayesian-coherence rows (shaded) share a reference posterior.

Insight 4. Acquisition is a belief problem. Agents lose by re-reading what they know rather than seeking what would change it. A testable belief names that evidence, so evidence acquisition will move from a fixed retrieval stage to a consequence of the belief.

8 The Test as a Metric: Evaluating Belief Apart from Outcome

A correct answer can follow from a correct belief, a wrong belief that did not matter, or a belief revised too late, and outcome metrics cannot separate the three, which is why stronger backbones do not yield stronger trading agents (Xie et al., 2024; Li et al., 2024a). Earlier agent benchmarks score task success or recall (Shridhar et al., 2020; Liu et al., 2023b; Yao et al., 2024). Recent work added belief-level metrics, each defining its own and each measuring one part from outside (calibration the belief state, accuracy the revision, Bayesian coherence the test, the belief–action gap whether the policy uses any). Fig. 5 aligns them by construct, and Table 14 in the appendix catalogues the benchmarks.

8.1 Constructs

Calibration over a trajectory has the most statistics and the least comparability. Its four formulations are never computed on another paper’s system (Lin et al., 2026c; Singh et al., 2026b; Dixon, 2026b; Ye et al., 2026), and the one replicated deficit flipped sign across model versions (Joshi, 2026). *Belief accuracy and revision* are scored against a known answer, a premise edit, human annotation, or a hidden ledger (Xu et al., 2026a; Dhanda, 2026; Myakala et al., 2026; Chao et al., 2026; Srivastava, 2026a), and *memory validity* per operation (Chen et al., 2025b; Hao et al., 2026). *Bayesian coherence* compares a belief trajectory with an ideal updater (Samanta et al., 2026; He

Table 6 | Credit levels compared by where their test comes from.

Level	Test from	Env.	No oracle	Belief	Act gap	Maturity
Output	humans, verifiers					
Trajectory	verifier, PRM					
Interaction	actions					
	memory ops					
State	task reward, teacher					
	oracle or teacher					
	closed world, teacher, target					
	own observation					
	returned observation					

Env., the test comes from the environment; No oracle, no external source of truth is needed; Belief, the written belief receives the gradient; Act gap, belief and action error are separated; Maturity, solid many systems, tint a few, grey none. Otherwise solid, yes; tint, partly; grey, no.

et al., 2025; Imran et al., 2025), and the *belief-action gap*, the youngest construct, scores belief and action on the same episodes (Deb and Krishnan, 2026; Gao et al., 2026b). Appendix D lists every instrument.

Every construct beyond calibration needs a latent state to compare against. Controlled environments provide it (Deb and Krishnan, 2026; Xu et al., 2026a; Alaswad et al., 2026; Wang et al., 2026i), long-horizon simulators know it but report belief through proxies (He et al., 2026b; Zhang et al., 2026c; Han et al., 2026), and historical replay provides neither (Peng et al., 2026b).

Insight 5. Belief cannot be scored by outcome. Seven constructs, more than twenty metrics, almost no cross-paper comparison, and only the fair oracle scores belief and action on the same episodes. Progress on belief will be measurable only where the latent state is known, so controlled environments, not replay, are where the next results should come from.

9 Discussion

Table 6 compares the credit levels by where their test comes from, and three observations follow.

From memory to testable belief: the test is scarce. The evidence supports the survey’s central claim that agents need to move from memory, which records what happened, to belief, a hypothesis their own observations can support or falsify. Memory has become a belief state and post-training a revision, but the step from record to hypothesis is the test, and Table 4 shows the state level taking it from a source the environment does not supply in every case but two, ReBel, which settles a predicate belief against the next observation, and ABEL, which reconstructs the previous one. Both are binary and isolated. The step of Fig. 2 that makes a belief a hypothesis is no longer empty, but it is thin. The cycle is broader than its Bayesian form and applies to agentic systems generally.

Auditability and expressiveness pull apart. Filters are validated in closed form but cannot in-

troduce a factor they were not given, written beliefs can name a new factor but are rarely testable, and world models are testable but mute. Hybrids that write the belief and check it parametrically are the convergence point (Song and Cai, 2026a; Zhang et al., 2026g; Kumar et al., 2026).

The belief-action gap is a training problem. Wherever belief and action are scored on the same episodes, prompting and oracles improve the belief while action-belief consistency barely moves (Deb and Krishnan, 2026; Gao et al., 2026b), and the system that moves both trains belief and policy jointly (Singh et al., 2026b). Closing the gap requires the state-level and interaction-level signals of Table 6 in one objective, not a better belief alone.

10 Open Problems

A calibrated belief with an environment-signed test. ReBel signs a written belief with the next observation, but a binary one, and Agent-BRACE carries certainty, but signed by a calibration target. Joining the two without the collapse dense prediction rewards show under group normalisation is the most direct open problem.

Latent beliefs without ground truth. Historical data has no oracle, closed world, or annotation. Filter-recovered latents and representation drift are candidate substitutes, unvalidated against each other (Dixon, 2026b; Xue, 2026), and a benchmark pairing a controlled environment with a replay of the same instrument would settle whether belief-level gains transfer.

Belief under governance. Regulated domains add auditability, which only fixed-space filters provide, and LLM-proposed spaces are the candidate bridge (Six et al., 2026; Soru, 2026; Chen et al., 2026c; Dixon, 2026c).

11 Conclusion

A memory records what has happened, whereas a belief is a hypothesis about what is true, open to testing and revision by future evidence. This survey argued that agents need the second, and asked of every system which of the belief’s three parts it has and where each comes from. The answer is uneven. The field has given the agent a state, from the trace to stores, filters, and world models, and a revision, with credit made denser at every level below the belief, and it has built more than

twenty ways to measure belief without a shared one. The test, the step that turns a record into a hypothesis, is still supplied from outside. The representations that are testable are not the ones the agent can reason about, the mechanisms that revise beliefs do not route the error to the claim that produced it, and the two methods that score the belief with the environment’s own returns score a binary belief and stand alone. The next step is a written belief that carries uncertainty, is scored by the agent’s own matured predictions, receives the error at the claim that made it, and is evaluated where the latent state is known. The taxonomy, screening record, and benchmark catalogue are released to make that step measurable.¹

Limitations

This survey covers work indexed by OpenAlex and arXiv from January 2020 to 30 September 2026, retrieved by the keyword protocol, citation-tree expansion, two targeted rounds, and the first monthly update (Appendix A); papers that describe belief maintenance without using any of our search terms may be missed. Our taxonomy assigns each method one source per part, although several systems combine sources; the codings were produced with large-language-model assistance under our codebook (Appendix A) and have not been independently replicated. Appendix A reports how each insight fares on the peer-reviewed subset alone. One directly relevant method reached us through a reader’s comment rather than the protocol; Appendix A reports the targeted search that followed. The survey reports numbers as stated by the cited papers and does not re-run any method; headline figures from preprints may change in later versions. Finance is used as the running example because delayed and partially observed evidence is the norm there; the taxonomy itself is domain-agnostic, but we have not validated it against embodied or dialogue agents in the same depth.

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¹<https://github.com/jiminHuang/belief-state-survey>

on Brev.

Declaration of interests

Jimin Huang and Xueqing Peng are affiliated with The Fin AI, which produced several of the finance benchmarks discussed here, among them FinBen and InvestorBench; these were screened, included, and described under the same protocol as every other paper (Appendix A). One benchmark discussed in Section 8 and Table 14, Herculean (Peng et al., 2026b), is co-authored by the authors of this survey. The authors declare no other financial or personal relationships with people or organisations that could be perceived to have influenced the work reported in this paper. The funder had no role in the design of the review, the screening protocol, the analysis, or the decision to publish.

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A Paper Collection Protocol

The search is scripted and rerunnable; the configuration (including every query string and term list), raw hits, per-stage counts, and the full-text coding record for the runs reported here (identifiers 20260915_v1 for 2025–2026 and 20260916_v3 for the 2020–2026 window) are released with the paper.

Queries. Twenty-three keyword queries were defined in the first run, four to five per survey section; a second run added eleven queries on evidence acquisition and theory-of-mind belief modelling (34 in total): representation (Section 4), revision (Section 5), learning (Section 6), evidence (Section 7), and evaluation (Section 8).

Sources. Source A queried OpenAlex across all venues with publication date from 1 January 2025, 50 results per query. Source B queried OpenAlex restricted to the arXiv source with publication date from 1 January 2026, 50 results per query, to improve recall of preprints. The arXiv API itself was rate-limited during the run; the OpenAlex mirror of arXiv lags by a few days.

Screening. Records were deduplicated by identifier and then by normalised title, so that a preprint and its journal version count once. Rule-based screening on title and abstract applied one inclusion criterion for an LLM-related term (I1), one exclusion criterion for off-domain applications (E1), and one inclusion criterion requiring at least two topic terms from a fixed list (I2). Records without an abstract of at least 200 characters were excluded. Included records were ranked by a relevance score used only for ordering the manual full-text screen, never for inclusion.

Table 7 | Paper-collection counts for run 20260915_v1.

Stage	Count
Queries	23
Source A raw hits / unique	1,150 / 715
Source B raw hits / unique	1,150 / 1,008
Merged unique (identifier)	1,685
After title-level deduplication	1,460
Excluded: no or short abstract	45
Excluded I1: no LLM term	160
Excluded E1: off-domain	35
Excluded I2: fewer than two topic terms	357
Included after rule screening	863
of which published in 2026	656
Full-text screened	117
of which included after reading	106

Second and third runs. The second run (34 queries, same sources and rules) returned 834 unique records, 338 after screening, of which 96 were published in 2026; only two of those had not already been read in the first run, which we take as evidence that the first run had saturated the keyword neighbourhood. The third run extended the window to 1 January 2020 with 100 results per query and returned 4,471 unique records, 3,999 after title deduplication and 2,079 after rule screening. Keyword ranking is dominated by generic high-citation papers before 2025, so the 2020–2025 foundations were enumerated as a curated list of 90 titles resolved through OpenAlex for

venue and citation data (Table 16); 75 of them were read in full and 73 included.

Citation-tree expansion. Keyword search cannot reach papers that describe belief maintenance in other words, so the included set was expanded by snowballing over the citation graph. Each round took the papers newly included in the previous round as seeds, fetched their reference lists (backward) and the papers citing them (forward) from the Semantic Scholar Graph API, kept candidates published from 2020, removed everything already screened, and applied the same rule-based screen as the keyword runs. Because forward citation of recent preprints is dense, the rule-included candidates were sampled for full-text reading by a fixed rule (topic score ≥ 10 , or the word *belief* in title or abstract, or linked to at least three seeds, or a world-model paper from 2025 with score ≥ 7), and every sampled paper was read and coded with the same fields as the keyword runs. Round one was seeded by the 179 keyword-included papers and each later round by the papers the previous round added; the full-text sample shrank from 302 to 115, 83, and 26 across four rounds, and the expansion stopped after the round whose sample fell below thirty, the stopping rule fixed in advance (Table 8).

Table 8 | Citation-tree expansion by round.

Stage	R1	R2	R3	R4
Seeds	178	165	55	45
Backward references	9,127	8,601	2,673	2,406
Forward citations	9,999	9,999	1,403	441
Unique candidates (2020–)	12,872	13,113	2,536	1,465
Already screened	302	1,284	605	465
Excluded by rules	8,448	7,401	1,300	747
Included after rule screening	4,122	4,428	631	253
Sampled for full-text reading	302	115	83	26
Included after reading	171	58	47	7
of which peer-reviewed	79	23	22	2

Forward citations are capped at 9,999 per request batch by the API.

Fifth round, targeted. After the four rounds above, a reader pointed us to a method the protocol had not reached (ReBel; Tang et al., 2026b), posted after most of our seeds and cited by almost no one yet. We treated it as a seed for a fifth, targeted round: the Semantic Scholar batch endpoint was rate-limited throughout, so we used ReBel’s own reference list (46 entries), an OpenAlex search on belief-consistency and belief-level supervision (85 candidates), and a Hugging

Face Papers search on nine related phrases (743 candidates, 211 after the rule screen and deduplication). Full-text screening added 23 papers (3 peer-reviewed), of which four score a belief or prediction against the agent’s own observations; every paper whose signal source mattered to a claim was re-read in full. One of the 23 (PABU) had passed the rule screen in the third keyword run but was never sampled, which is the protocol’s own miss. The additions are marked R5 in the released record and in Table 9.

Sixth round. A final targeted round added four papers identified by the authors and readers after the fifth round: two posted after the search cut-off, Jev-Mem and REFLEX (Jiang et al., 2026a; Wu and Lim, 2026), and two the keyword and citation searches did not reach, FLAG-TRADER and Conv-FinRe (Xiong et al., 2025a; Wang et al., 2026i). They were read in full, coded with the same fields, and are counted in the totals, the sensitivity analysis, and the master table (column R6 of Table 9).

First monthly update. The released record is refreshed monthly with the same OpenAlex and Hugging Face Papers queries and rule screen, restricted to the new window. The September 2026 window returned 1,051 unique records and 434 rule-screened candidates, of which 47 were read in full and 29 included (3 peer-reviewed): among them two surveys that formalise a Bayes-filter belief for one setting (Cai et al., 2026; Tang et al., 2026c) and three methods that credit a probabilistic belief against a known answer (Zhang et al., 2026j; Wang et al., 2026e; Hu et al., 2026a). They are counted throughout (column U1 of Table 9), and no insight changes direction (Table 10).

Year distribution. Table 9 gives the included papers by year, source, and whether peer-reviewed. The share that is peer-reviewed falls with recency, from four in five before 2025 to about one in nine for 2026, which is the ordinary lag of review rather than a property of the field.

Evidence level. Of the 518 papers included after full-text reading (179 from keyword search, 283 from citation-tree expansion, 23 from the targeted fifth round, 4 from the sixth, and 29 from the first monthly update), 181 are peer-reviewed and 337 are preprints; every number quoted in the text is the paper’s own headline figure, taken from the

Table 9 | Included papers by publication year and by the run that found them (keyword search, citation-tree or targeted round, or monthly update).

Year	Keyword	R1	R2	R3	R4	R5	R6	U1	Total	Peer-rev.
2020		1	3	1	0	0	0	0	5	4
2021		2	2	1	0	0	0	0	5	3
2022		9	4	0	1	0	0	0	14	13
2023		27	21	2	0	0	1	0	51	41
2024		16	20	9	4	1	1	0	51	40
2025		18	69	20	20	2	5	1	135	51
2026		106	52	25	22	4	16	3	257	29
Total		179	171	58	47	7	23	4	518	181

abstract or the reported table, and none has been reproduced by us.

Full-text screening. A paper passed full-text screening if it (a) concerns a language-model agent or a component used by one, and (b) constructs, revises, tests, learns, or evaluates an explicit state carried between decisions, or documents a failure of such a state; benchmarks and surveys that do (b) were kept and coded as such. Each paper read in full was recorded with the fields: who maintains the belief (source of the belief state), credited object, revision signal (source of the revision and test), whether belief-level metrics are reported, survey section, and role. The record is the source of Fig. 1, Tables 11–14, and Fig. 5. Screening and coding were carried out with large-language-model agents: each sampled paper was read and coded against these fields under the fixed codebook above, and every paper whose coding supports a claim in the text was re-read in full. The authors defined the framework, the inclusion criteria, and the codebook, and are responsible for the record; it has not been re-coded by a second annotator.

Peer-reviewed subset. Because 337 of the 518 papers are preprints, Table 10 recomputes the counts behind each insight box on the 181 peer-reviewed papers alone.

B Additional Tables and Figures

C Timeline of Learning Signals by Credited Object

D Details for Sections 6 and 8

State-level learning signals (Section 6.3). Until 2026 every such method took its test from outside the environment. CBM trains with belief-state rewards from symbolic verification and cuts belief-

Table 10 | Sensitivity of the five insights to preprint reliance: the counts each rests on, over all included papers and over the peer-reviewed subset alone.

#	Counts the insight rests on	All	P only
1	belief state written / filter / world model	71 / 27 / 55	22 / 8 / 19
2	revision papers (§5)	122	37
3	credited: interaction / state / state by own obs.	95 / 50 / 1	33 / 13 / 0
4	acquisition papers (§7)	66	18
5	papers with a belief-level metric	105	34

Every insight is a claim about a proportion or an absence, and each holds in the same direction on the peer-reviewed subset; the cell for state credit signed by the agent’s own observation holds one paper, a preprint, in both.

management failures by about 71% (Xu et al., 2026a), Agent-BRACE trains belief model and policy jointly with verbalised certainty as the belief’s output (Singh et al., 2026b), and BOND distils a Bayesian teacher’s posterior into an 8B student (Cui and Lin, 2026). VAGEN rewards a written state prediction against the simulator’s state (Wang et al., 2025d), T3 truncates rollouts by a belief deviation anchored to the true hypothesis (Zou et al., 2025), IGPO and InfoPO reward a turn by the belief update it causes toward the known answer (Wang et al., 2025a; Kong et al., 2026), Dyna-Think, SPA, and RLVR-World train state prediction as a loss or verifiable reward (Yu et al., 2025c; Chen et al., 2025e; Wu et al., 2025a), DEL-ToM scores each belief-update step with a process belief model (Wu et al., 2025d), and CORE and GAVEL train a probabilistic belief against gold slots or held-out placements (Zhang et al., 2026j; Wang et al., 2026e). Each needs the answer or the latent state at training time. Three systems now take it from the agent’s own observations: ReBel rewards each predicate of a written belief that the next observation supports, with the gradient reaching the belief (Tang et al., 2026b); ABBEL grades a natural-language belief by how well the prior observation can be reconstructed from it (Lidayan et al., 2025); PaW adds a next-observation loss on the policy (Lu et al., 2026a). None carries uncertainty, and the dense form is fragile: the same reward through group-normalised advantages collapses every run, and as an auxiliary loss it is matched by shuffled targets (Wang, 2026a).

Belief-level constructs (Section 8.1). *Calibration over a trajectory* has the most statistics and the least comparability: checkpoint and episode calibration, posterior entropy, and proper scoring rules on time-gated forecasts express one construct four ways, none computed on another pa-

Table 11 | The belief state by who maintains it, with the revision and test each system carries.

Class	Method	State written by model	Append-only	Uncertainty	Test (TEST)	Revision (REVISE)	Domain
Context as state	ReAct (Yao et al., 2023b)	✓	✓	✗	✗	✗	general
External memory	MemGPT (Packer et al., 2023)	✗	✗	✗	✗	store update	dialogue
Probabilistic memory	BeliefMem (Liao et al., 2026a)	✗	✗	prob.	✗	noisy-OR	LoCoMo, ALFWorld
	D-MEM (Song and Xin, 2026)	✗	✗	RPE	✗	gated write	LoCoMo
External filter	Belief-State Engine (Chattopadhyay and Halder, 2026)	✗	✗	posterior	✓	Bayes	Tiger, security
	Belief at Risk (Dixon, 2026b)	✗	✗	posterior	✓	Bayes	equities
Model-written	Agent-BRACE (Singh et al., 2026b)	✓	✓	ordinal	✗	RL	ALFWorld-style
	Linguistic forecaster (Murphy, 2026)	✓	✓	numeric	✓	sequential	ForecastBench
	BOND (Cui and Lin, 2026)	✓	✓	posterior tag	✗	distilled	negotiation
	CBM (Xu et al., 2026a)	✓	✓	✗	✓ (symbolic)	RL	closed world
	EvoSCM (Zhao et al., 2026b)	✓	✓	population	✓ (interventions)	rule distillation	science
	ReBel (Tang et al., 2026b)	✓	✗	✗	✓ (next observation)	RL	ALFWorld, WebShop
	ABBEL (Lidayan et al., 2025)	✓	✗	✗	reconstruction	RL	text games, ColBench
	World model	✗	✗	latent	✓	learned	general
	BB-WM (Kumar et al., 2026)	✗	✗	latent	✓	learned	general

TEST: the belief commits to a testable prediction.

Table 12 | Belief revision methods and benchmarks by failure mode addressed, revision trigger, and where the signal is projected.

Method	Failed stay	Failed update	Failed isolation	Failed act	Trigger	Signal projected to
Reflexion (Shinn et al., 2023)		✓			task failure	whole context
BeliefShift (Myakala et al., 2026)	✓	✓			user evidence	(benchmark)
STALE (Chao et al., 2026)		✓		✓	implicit conflict	(benchmark)
CBM (Xu et al., 2026a)	✓	✓	✓		symbolic check	belief state
TOKI (Wang, 2026b)		✓			contradiction	memory rows
MemOps (Hao et al., 2026)		✓	✓		lifecycle op	(benchmark)
D-MEM (Song and Xin, 2026)		✓	✓		reward pred. error	memory path
STOCKTAKE (Deb and Krishnan, 2026)		✓		✓	oracle mismatch	(benchmark)
Agent-BRACE (Singh et al., 2026b)		✓		✓	certainty	belief model
Werewolf audit (Gao et al., 2026b)		✓		✓	external belief log	(benchmark)

per’s system (Lin et al., 2026c; Singh et al., 2026b; Chattopadhyay and Halder, 2026; Dixon, 2026b; Ghosh and Devarakonda, 2026; Ye et al., 2026); one replicated deficit flipped sign across model versions (Joshi, 2026). *Belief accuracy and revision* are measured against a known answer in a closed world, after a minimal premise edit, and across recall, inference, and change detection (Xu et al., 2026a; Dhanda, 2026; Nguyen et al., 2026), against human annotation or a hidden ledger (Myakala et al., 2026; Chao et al., 2026; Hao et al., 2026; Yang et al., 2026b; Srivastava, 2026a), a probed spatial map or a screenplay’s state (Zhang et al., 2026f; Tian et al., 2026), and theory-of-mind benchmarks (Kim et al., 2023b; Bawatneh et al., 2026; Wang et al., 2026c). *Memory validity* is scored per operation or by penalising invalidated memories (Chen et al., 2025b; Wu et al., 2024; Liu et al., 2026a; Uddin et al., 2026). *Bayesian coherence* compares a belief trajectory with an ideal updater: total-variation distance to the exact posterior (Samanta et al., 2026), a martingale test on successive beliefs (He et al., 2025), and the correlation between the model’s update and the log-odds of the evidence (Imran et al., 2025; Yuan et al., 2025b). The *belief–action gap* is the youngest construct: STOCKTAKE scores the agent against an exact Bayes filter on the same

observations (Deb and Krishnan, 2026), the Werewolf audit reports action–belief consistency (Gao et al., 2026b), and SimpleToM separates inferring a belief from applying it (Gu et al., 2024).

Table 13 | Learning signals by credited object, with the source of the test that signs them.

Level	Representative methods	Test from	Granularity	Signed by returned σ_{t+1}	Delayed evidence	Partial obs.
Output	SFT, RLHF, DPO (Ouyang et al., 2022; Rafailov et al., 2023)	demos, preferences	response	$\times$	$\times$	$\times$
Trajectory	PPO, GRPO, PRMs (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024)	verifier, PRM	step / rollout	$\times$	$\times$	$\times$
Interaction	HiPER, G ² -PO, ECPO, IAPO (Peng et al., 2026a; Wang et al., 2026; Li et al., 2026; Ren et al., 2026a)	environment	turn / action	$\times$	partial	$\times$
State	Memory-R2, TAMTRL, EvoHarness-RL (Yan et al., 2026a; Wang et al., 2026d; Ning et al., 2026)	environment, teacher	memory operation	$\times$	partial	$\checkmark$
	CBM (Xu et al., 2026a)	symbolic belief reward	belief state	closed world	$\checkmark$	$\checkmark$
	Agent-BRACE (Singh et al., 2026b)	joint RL, calibration	claim + certainty	calibration target	$\checkmark$	$\checkmark$
	BOND (Cui and Lin, 2026)	Bayesian teacher	posterior tag	teacher	$\times$	$\checkmark$
	CORE (Zhang et al., 2026j)	gold slot values	probabilistic slot belief	known answer	$\times$	$\checkmark$
	GAVEL (Wang et al., 2026e)	held-out placements	posterior over object locations	known answer	$\times$	$\checkmark$
	ReBel (Tang et al., 2026b)	own next observation	predicate belief		$\checkmark$	$\checkmark$
	ABBEL (Lidayan et al., 2025)	reconstruction of prior observation	natural-language belief	$\times$ (backward)	$\checkmark$	$\checkmark$
	PaW (Lu et al., 2026a)	own next observation	prediction on the policy	$\checkmark$	$\times$	$\checkmark$

Table 14 | Benchmarks that exercise belief maintenance, by whether the evaluator can observe the latent state and what belief-level quantity is reported.

Benchmark	Domain	Horizon	Latent state	Observable to evaluator	Delayed evidence	Belief-level quantity reported
ALFWorld (Shridhar et al., 2020)	text + embodied	episodic	household state	$\checkmark$	partial	none (task success)
PlanBench (Valmeekam et al., 2022)	classical planning	plan-length	PDDL state	$\checkmark$	$\times$	plan verification, replanning
AgentBench (Liu et al., 2023b)	8 environments	episodic	environment state	$\checkmark$	partial	none (task success)
SmartPlay (Wu et al., 2023)	6 games	episodic	game state	$\checkmark$	partial	none (9 capability scores)
T4D / FaR (Zhou et al., 2023b)	theory of mind	single-turn	others' beliefs	$\checkmark$ (annotated)	$\times$	belief tracking vs. action selection
False-belief tasks (Kosinski, 2023)	theory of mind	single-turn	others' beliefs	$\checkmark$ (annotated)	$\times$	false-belief accuracy
LoCoMo (Maharana et al., 2024)	long conversations	35 sessions	conversation facts	annotated	$\checkmark$	temporal QA (proxy)
τ -bench / τ^2 -bench (Yao et al., 2024; Barres et al., 2025)	tool-user interaction	multi-turn	database state	$\checkmark$	$\checkmark$	pass*k consistency; reasoning vs. communication errors
MemoryAgentBench (Hu et al., 2025c)	incremental multi-turn	long-horizon	memory contents	annotated	$\checkmark$	retrieval, test-time learning, conflict resolution
BeliefTrack (Xu et al., 2026a)	closed-world reasoning	multi-turn	finite belief space	$\checkmark$ (symbolic)	$\checkmark$	turn-level belief accuracy; failed stay/update/isolation
DeltaLogic (Dhanda, 2026)	logical reasoning	2 steps	premise set	$\checkmark$	$\checkmark$	revision accuracy vs. initial accuracy
DTOM-Track (Nguyen et al., 2026)	dialogue	multi-turn	interlocutor belief	$\checkmark$ (annotated)	$\checkmark$	prior-belief recall; change detection
BeliefShift (Myakala et al., 2026)	user modelling	multi-session	user opinion	annotated	$\checkmark$	revision accuracy; drift coherence; evidence sensitivity
OmniToM (Bawatneh et al., 2026)	theory of mind	narrative	actors' beliefs	$\checkmark$ (annotated)	$\times$	belief-proposition extraction; schema labels
MINDGAMES (Wang et al., 2026c)	social games	repeated games	hidden roles, opponents	$\checkmark$	$\checkmark$	TrueSkill; turn-level logs
MEDLEY-BENCH (Abtahi et al., 2026)	social pressure	2 rounds	—	$\times$	$\times$	sensitivity to manipulated consensus
STALE (Chao et al., 2026)	memory validity	multi-turn	fact validity	annotated	$\checkmark$	state resolution; premise resistance; policy adaptation
MemOps (Hao et al., 2026)	long conversations	multi-session	memory lifecycle	annotated	$\checkmark$	operation-level probes
GroupMemBench (Yang et al., 2026b)	multi-party dialogue	multi-session	per-speaker knowledge	annotated	$\checkmark$	information-update accuracy
MemGym (Xu et al., 2026c)	mixed agent tasks	long-horizon	—	$\times$	partial	memory-isolated score delta
STOCKTAKE (Deb and Krishnan, 2026)	inventory control	weekly, long	demand factors	$\checkmark$ (Bayes filter)	$\checkmark$	detection lag; skill vs. oracle; knowing-doing rate
Flux (Alaswad et al., 2026)	text game	long-horizon	simulator state	$\checkmark$	$\checkmark$	state-tracking error modes
YC-Bench (He et al., 2026b)	startup simulation	hundreds of turns	market, clients	privileged outcome	$\checkmark$	scratchpad usage (proxy)
RetailBench (Zhang et al., 2026c)	retail POMDP	180 days	demand, supply	privileged oracle	$\checkmark$	evidence-acquisition behaviour (proxy)
EnterpriseArena (Han et al., 2026)	lending	132 months	borrower pool	privileged outcome	$\checkmark$	observation-budget allocation (proxy)
Business Arena (Pan et al., 2026a)	marketplace	long-horizon	market	privileged outcome	$\checkmark$	action-level attribution
Werewolf audit (Gao et al., 2026b)	social deduction	9-player game	roles	$\checkmark$	$\checkmark$	action-belief consistency
CAGE-2 study (Bogdanov et al., 2026)	cyber defence	episodic	network state	$\checkmark$	partial	state-abstraction return gain
MD5 state tracking (Pai and Xian, 2026)	tool chains	196 calls	registers	$\checkmark$	$\times$	exact-state success
FinBench (time-gated) (Ghosh and Devarakonda, 2026)	asset forecasting	daily	returns	realised	$\checkmark$	Brier; interval scores
TradeArena (Xue, 2026)	trading	multi-week	regimes	realised	$\checkmark$	representation drift (early warning)
HERCULEAN (Peng et al., 2026b)	finance workflows	multi-step	market, books	partial	$\checkmark$	state-tracking sub-score
FinalityBench (Sharma, 2026)	financial operations	multi-step	canonical event log	$\checkmark$	$\checkmark$	none (effect-level loss)
Forecast-Dojo (Ye et al., 2026)	event forecasting	dated timeline	event outcomes	$\checkmark$ (resolution)	$\checkmark$	per-step Brier, log loss
EnterpriseBench (Yang et al., 2026d)	enterprise decisions	interactive	hidden information	privileged	$\checkmark$	none (task level)
Ledger memory benchmark (Srivastava, 2026a)	memory	multi-session	principals' beliefs	$\checkmark$ (ledger)	$\checkmark$	expected belief state per principal
Werewolf accusations (Yang et al., 2026f)	social deduction	single message	player roles	$\checkmark$	$\times$	prior suspicion and shift per accusation
Regent Chess (Joshi, 2026)	board game	game	hidden royal piece	$\checkmark$	$\checkmark$	action-time Brier
FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024a)	finance tasks / trading	task / episodic	—	$\times$	partial	none (outcome only)

"Annotated" means human labels stand in for the latent state; "privileged" means the simulator knows it but the benchmark reports only outcomes.

Table 15 | Systems with an explicit belief mechanism, by year.

System	Year	State kept by	Credited	Revision trigger	Metric
Cyclic subtask graphs (Gharzaddine and Saab, 2026)	2026	written	none	state-analysis-and-routing...	
CHAL (Giovannelli and Kent, 2026)	2026	written	none	gradient-informed revision on...	o
Pinductor (Six et al., 2026)	2026	world model	world-model loss	belief-based likelihood score...	o
Belief Engine stance... (Yang et al., 2026c)	2026	prob. memory	none	log-odds update with evidence...	$\checkmark$
HiPF (Diao et al., 2026)	2026	written	trajectory	subgoal completion +...	
ProFrag (Ma et al., 2026b)	2026	world model	trajectory	post-episode graph refinement...	o
POMDP validation... (Dixon, 2026c)	2026	ext. filter	none	Bayesian	$\checkmark$
Internalizing the Future (Zhang et al., 2026h)	2026	written	trajectory	foresight-conditioned RL...	o
Hybrid-WM (Song and Cai, 2026a)	2026	world model	world-model loss	consistency gate	$\checkmark$
Governance-aware POMDP... (Dixon, 2026a)	2026	ext. filter	none	Bayesian update on evidence...	o
BayesEvolve (Wu et al., 2026d)	2026	ext. filter	none	posterior refit after each...	$\checkmark$
WorldEvolver (Zhang et al., 2026g)	2026	world model	world-model loss	prediction-observation...	o
WM-SAR (Song and Cai, 2026b)	2026	ext. filter	trajectory	spectral error amplification	
BCO (Chen et al., 2026c)	2026	written	none	evaluation of new candidate...	$\checkmark$
EvosCM causal belief... (Zhao et al., 2026b)	2026	written	none	prediction-observation...	$\checkmark$
CAPTURE (Hossain et al., 2026)	2026	world model	world-model loss	uncertainty-triggered...	$\checkmark$
Semantic Bayesian World... (Soru, 2026)	2026	ext. filter	none	Bayesian conditioning on...	
Belief-State Engine (Chattopadhyay and Halder, 2026)	2026	ext. filter	none (prompting)	Bayesian posterior	$\checkmark$
ReBel (Tang et al., 2026b)	2026	written	state	next observation (consistency)	o
ABBEL (Lidayan et al., 2025)	2026	written	state	reconstruction of prior obs.	o
PABU (Uyang et al., 2026c)	2026	written	state	teacher progress labels	o
MARBO (Hwang et al., 2026)	2026	written	state	ground-truth roles	o
CORE (Zhang et al., 2026i)	2026	prob. memory	state	gated slot revision	$\checkmark$
GAVEL (Wang et al., 2026e)	2026	ext. filter	state	online Bayesian update	$\checkmark$
BCA (Akbar et al., 2026)	2026	ext. filter	none	Beta update on appraised...	$\checkmark$
MoM / P-Mem (Qin and Lu, 2026)	2026	written	none	typed write-time operations	$\checkmark$

State kept by: who maintains the belief (Section 4); Credited: the object that receives the learning signal under the criterion of Section 6, *none* if the belief is not trained; Metric: $\checkmark$ reports a belief-level quantity, o a proxy.

Table 16 | Every paper included after full-text reading, ordered by year and first author.

System	Year	State kept by	Credited	Revision trigger	Metric
GATA belief graphs (Adhikari et al., 2020)	2020	world model	state	graph updater infers and updates latent. . .	○
Text games with common sense (Dambekodi et al., 2020)	2020	ext. memory	trajectory	commonsense inference (COMET/BERT) fill. . .	○
RAG (Lewis et al., 2020)	2020	ext. memory	output	retrieved passages condition generation. . .	
ToM and decision science typology (Rusch et al., 2020)	2020	—	—	none	
ALFWorld (Shridhar et al., 2020)	2020	—	output	n/a (environment)	
Decision Transformer (Chen et al., 2021)	2021	context	interaction	return-to-go conditioning over past sta. . .	
Do LMs have beliefs (Hase et al., 2021)	2021	—	state	belief-update objective (SLAG) applied. . .	✓
BeliefBank (Kassner et al., 2021)	2021	ext. memory	—	weighted MaxSAT revises beliefs that cl. . .	✓
Schema-driven DST (Lee et al., 2021)	2021	written	state	dialogue history plus schema descriptio. . .	✓
WebGPT (Nakano et al., 2021)	2021	context	trajectory	none (browsing history in context; rewa. . .	
SayCan (Ahn et al., 2022)	2022	—	interaction	affordance value functions score skill. . .	
LMs as Agent Models (Andreas, 2022)	2022	—	—	n/a (position paper)	
Towards Teachable Reasoning Syste. . . (Mishra et al., 2022)	2022	ext. memory	—	user corrections of erroneous model bel. . .	○
Inner Monologue (Huang et al., 2022b)	2022	context	interaction	success detection, scene description, h. . .	
LLMs Can Self-Improve (Huang et al., 2022a)	2022	—	output	self-consistency majority vote as pseud. . .	
LMs (Mostly) Know (Kadavath et al., 2022)	2022	written	output	self-evaluation P(True) and P(IK) of ow. . .	✓
Verbalized Uncertainty (Lin et al., 2022)	2022	written	output	fine-tuning on calibrated verbal confid. . .	✓
HELM (Paischer et al., 2022)	2022	ext. memory	trajectory	frozen LM compresses observation histor. . .	
ILQL (Snell et al., 2022)	2022	context	trajectory	none (offline returns from logged dialo. . .	
PlanBench (Valmeekam et al., 2022)	2022	—	output	n/a (benchmark; IPC planning domains)	
Self-Consistency (Wang et al., 2022)	2022	—	output	marginalisation over sampled reasoning. . .	
Chain-of-Thought (Wei et al., 2022b)	2022	context	output	none (single forward pass; intermediate. . .	
WebShop (Yao et al., 2022a)	2022	context	—	none (environment; partial observabilit. . .	
ReAct (Yao et al., 2022b)	2022	context	—	reasoning traces used to induce, track. . .	
Self-RAG (Asai et al., 2023)	2023	context	output	self-assessed retrieval need (reflectio. . .	○
NLMs as epistemic agents (Betz et al., 2023)	2023	—	—	novel evidence item propagated through. . .	✓
Self-Debug (Chen et al., 2023b)	2023	context	output	execution feedback + self-generated cod. . .	
FireAct (Chen et al., 2023a)	2023	context	trajectory	supervised fine-tuning on GPT-4 ReAct/C. . .	
DreamerV3 (Hafner et al., 2023)	2023	world model	state	RSSM latent posterior updated from obse. . .	
RAP (Hao et al., 2023)	2023	world model	trajectory	LLM-as-world-model predicted next state. . .	
HI-TOM (He et al., 2023)	2023	context	—	none (story-based false-belief question. . .	✓
LAW (Hu and Shu, 2023)	2023	world model	—	position paper: world and agent models. . .	
RCI (Kim et al., 2023a)	2023	context	interaction	recursive self-criticism of proposed ac. . .	
FANToM (Kim et al., 2023b)	2023	context	—	none (evaluation)	✓
LLM ToM (Kosinski) (Kosinski, 2023)	2023	—	output	n/a (probing)	✓
Semantic Entropy (Kuhn et al., 2023)	2023	—	output	entropy over meaning-clustered samples	✓
RecallIM (Kynoch et al., 2023)	2023	ext. memory	memory op	new knowledge statement triggers concep. . .	✓
TradingGPT (Li et al., 2023b)	2023	ext. memory	interaction	layered memory decay + inter-agent deba. . .	
ToM Multi-Agent Collab (Li et al., 2023a)	2023	written	—	prompted explicit belief-state update e. . .	✓
Let's Verify Step by Step (Lightman et al., 2024)	2023	—	output	per-step human process labels (PRM800K)	
AgentBench (Liu et al., 2023b)	2023	—	output	n/a (benchmark)	
LLM+P (Liu et al., 2023a)	2023	—	output	n/a (translate to PDDL, symbolic planne. . .	
RAFA (Liu et al., 2023c)	2023	context	—	switching condition: replan when predic. . .	○
Self-Refine (Madaan et al., 2023)	2023	context	output	self-generated natural-language feedbac. . .	
RET-LLM (Modarressi et al., 2023)	2023	ext. memory	—	LLM emits memory-write of extracted tri. . .	
DECKARD (Nottingham et al., 2023)	2023	world model	WM loss	LLM-hypothesized abstract world model v. . .	○
MemGPT (Packer et al., 2023)	2023	ext. memory	output	self-directed memory paging via functio. . .	
Generative Agents (Park et al., 2023)	2023	ext. memory	interaction	observation stream; periodic reflection. . .	
DPO (Rafailov et al., 2023)	2023	—	output	none	
SymbolicToM (Sclar et al., 2023)	2023	ext. filter	—	symbolic graph update on each narrated. . .	✓
Clever Hans N-ToM (Shapira et al., 2023)	2023	context	—	none (evaluation)	○
Sycophancy (Sharma et al., 2023)	2023	context	output	user pushback / stated user view (socia. . .	○

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 17 | Master table, continued (2/10).

System	Year	State kept by	Credited	Revision trigger	Metric
Reflexion (Shinn et al., 2023)	2023	ext. memory	trajectory	verbal self-reflection after episode fa...	
CoALA (Sumers et al., 2023)	2023	ext. memory	–	conceptual: learning actions update mem...	
Interactive planning POMDP (Sun et al., 2023)	2023	context	–	none	
LLaRP (Szot et al., 2023)	2023	context	trajectory	none (egocentric observations in contex...	
Voyager (Wang et al., 2023a)	2023	ext. memory	trajectory	environment execution feedback, errors,...	
DEPS (Wang et al., 2023c)	2023	context	trajectory	descriptor of execution outcome; self-e...	
Math-Shepherd (Wang et al., 2023b)	2023	–	output	automatic per-step labels via MC comple...	
SimToM (Wilf et al., 2023)	2023	context	–	perspective-taking filter: context rest...	○
SmartPlay (Wu et al., 2023)	2023	–	output	n/a (benchmark; game environments)	
Tree of Thoughts (Yao et al., 2023a)	2023	context	trajectory	self-evaluation of intermediate thought...	
FinMem (Yu et al., 2023)	2023	ext. memory	interaction	layered memory with recency/relevancy/i...	
AgentTuning (Zeng et al., 2023)	2023	context	trajectory	hybrid instruction tuning on AgentInstr...	
CoELA (Zhang et al., 2023b)	2023	ext. memory	trajectory	semantic memory of world/agents/tasks u...	
ProAgent (Zhang et al., 2023a)	2023	written	–	teammate intention inferred from observ...	○
LLM-MCTS (Zhao et al., 2023b)	2023	world model	trajectory	LLM-sampled world state belief (object...	
ExpeL (Zhao et al., 2023a)	2023	ext. memory	trajectory	cross-task insight extraction from succ...	
Synapse (Zheng et al., 2023)	2023	ext. memory	–	none (exemplar retrieval by similarity)	
MemoryBank (Zhong et al., 2023a)	2023	ext. memory	output	Ebbinghaus-style forgetting with time d...	
MQuAKE / MeLLO (Zhong et al., 2023b)	2023	ext. memory	–	edited fact stored externally; iterativ...	✓
FaR / T4D (Zhou et al., 2023b)	2023	context	interaction	foresee-and-reflect structured promptin...	
RecurrentGPT (Zhou et al., 2023d)	2023	written	output	LLM rewrites its own natural-language s...	
LATS (Zhou et al., 2023a)	2023	ext. memory	trajectory	MCTS backup of LM-valued rewards + self...	
WebArena (Zhou et al., 2023c)	2023	context	–	none	○
DigiRL (Bai et al., 2024)	2024	context	interaction	doubly-robust step-level advantage mixi...	
Latent state estimation for UI ag... (Bishop et al., 2024)	2024	written	–	none	
WMA Web Agent (Chae et al., 2024)	2024	world model	WM loss	predicted next-observation as free-form...	○
NegotiationToM (Chan et al., 2024)	2024	–	–	none	✓
ToMBench (Chen et al., 2024b)	2024	–	–	none	○
Agent-FLAN (Chen et al., 2024a)	2024	context	trajectory	none (SFT on decomposed agent corpora;...	○
Talker-Reasoner (Christakopoulou and Mourad, 2024)	2024	written	state	Reasoner (System 2) writes/updates user...	
Hypothetical Minds (Cross et al., 2024)	2024	written	state	hypotheses about other agents’ strategi...	
GIF-MCTS code world models (Dainese et al., 2024)	2024	world model	WM loss	unit-test and trajectory feedback drive...	○
InferAct (Fang et al., 2024)	2024	written	interaction	ToM-style belief inference over the age...	○
SimpleToM (Gu et al., 2024)	2024	–	–	none	✓
Rational belief revision in model... (Hase et al., 2024)	2024	–	–	model edits evaluated against an ideali...	✓
From Words to Actions (He et al., 2024)	2024	context	–	in-context Bayesian aggregated imitatio...	
Coherence norms (Hofweber et al., 2024)	2024	–	–	coherence norms on credence (next-token...	
TimeToM (Hou et al., 2024)	2024	written	–	temporal ordering of events; belief com...	○
HiAgent (Hu et al., 2024)	2024	written	–	subgoal completion judged by the LLM tr...	○
MMToM-QA (Jin et al., 2024)	2024	prob. memory	–	Bayesian inverse planning updates poste...	✓
PercepToM (Jung et al., 2024)	2024	written	–	perception inference feeds a perception...	✓
QuBE (Kim et al., 2024)	2024	written	state	question-answering over observations to...	
VisualWebArena (Koh et al., 2024)	2024	–	output	n/a (benchmark)	
SCoRe (Kumar et al., 2024)	2024	context	interaction	multi-turn online RL with a reward bonu...	
ReadAgent (Lee et al., 2024)	2024	ext. memory	–	agent decides episode boundaries, compr...	
Embodied Agent Interface (Li et al., 2024b)	2024	world model	–	none (evaluation only)	○
MindForge (Lic et al., 2024)	2024	written	–	new percepts and inter-agent messages u...	
LoCoMo (Maharana et al., 2024)	2024	–	output	n/a (benchmark; QA/event summarization...	
Agent Q (Putta et al., 2024)	2024	ext. memory	trajectory	MCTS with AI self-critique feedback + o...	
WebRL (Qi et al., 2024)	2024	–	output	outcome-supervised reward model; self-g...	
r to Q* (Rafailov et al., 2024)	2024	–	output	none (token-level implicit reward)	
Belief state geometry (Shai et al., 2024)	2024	context	–	implicit Bayesian belief updating over...	✓

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 18 | Master table, continued (3/10).

System	Year	State kept by	Credited	Revision trigger	Metric
MuMA-ToM (Shi et al., 2024)	2024	world model	–	Bayesian inverse multi-agent planning o...	✓
FortUne Dial (Sicilia et al., 2024)	2024	context	output	fine-tuning (supervised and off-policy...	✓
Common-ToM (Soubki et al., 2024)	2024	written	–	explicit belief representation supplied...	○
self-verification limits (Stechly et al., 2024)	2024	context	–	self-critique vs sound external verifie...	
AppWorld (Trivedi et al., 2024)	2024	context	–	none (environment; state-based unit tes...	○
SOTOPIA-pi (Wang et al., 2024a)	2024	–	output	LLM-rated social goal completion used f...	
Agent Workflow Memory (Wang et al., 2024d)	2024	ext. memory	–	induction of reusable workflows from su...	
DistRL (Wang et al., 2024b)	2024	context	trajectory	none (state is screenshot history in co...	
MEMORYLLM (Wang et al., 2024c)	2024	prob. memory	state	new text knowledge injected into latent...	○
POAD (Wen et al., 2024)	2024	context	interaction	token-level Bellman backup (BAD) assign...	
Belief-R (Wilie et al., 2024)	2024	context	–	new premise that should defeat prior in...	✓
LongMemEval (Wu et al., 2024)	2024	ext. memory	–	knowledge-update questions test whether...	✓
AgentGym (Xi et al., 2024)	2024	context	trajectory	AgentEvol: iterative exploration + SFT...	
TradingAgents (Xiao et al., 2024)	2024	context	output	multi-agent debate (bull vs bear resear...	
tau-bench (Yao et al., 2024)	2024	–	–	simulated-user utterances and API tool...	
LaBToM (Ying et al., 2024)	2024	ext. filter	–	Bayesian inverse planning over the obse...	✓
FinCon (Yu et al., 2024)	2024	ext. memory	state	episodic self-critique producing concep...	
FinAgent (Zhang et al., 2024a)	2024	ext. memory	interaction	low-level and high-level reflection ove...	
MemSim (Zhang et al., 2024c)	2024	ext. memory	–	none (evaluation simulator)	○
ArCHer (Zhou et al., 2024b)	2024	–	interaction	hierarchical RL: utterance-level off-po...	
Reflect-RL (Zhou et al., 2024a)	2024	written	interaction	frozen reflection model produces natura...	
Beliefs of self and others (Zhu et al., 2024)	2024	context	–	none (probing and activation interventi...	✓
Bayesian geometry of attention (Agarwal et al., 2025)	2025	context	–	implicit posterior update via FFN in HM...	✓
Markovian Thinker (Aghajohari et al., 2025)	2025	written	output	none	
MemoryBench (Ai et al., 2025)	2025	ext. memory	–	accumulated simulated user feedback dur...	○
BASIL (Atwell et al., 2025)	2025	context	output	user pushback vs evidence; Bayesian-con...	✓
Vending-Bench (Backlund and Petersson, 2025)	2025	ext. memory	–	none (evaluation only; agent has scratc...	○
Align While Search (Bae et al., 2025)	2025	ext. filter	–	posterior update from action-conditione...	✓
tau2-bench (Barres et al., 2025)	2025	–	–	user actions on shared environment obse...	
LOOP (Chen et al., 2025c)	2025	context	trajectory	environment feedback via API observatio...	
ReSearch (Chen et al., 2025d)	2025	context	output	search results inserted into reasoning...	
HaluMem (Chen et al., 2025b)	2025	ext. memory	–	memory extraction and memory updating o...	✓
SPA (Chen et al., 2025e)	2025	world model	WM loss	self-experience SFT supervises current...	○
GUI-Shepherd (Chen et al., 2025a)	2025	–	interaction	process reward model trained on 52k hum...	
Agent-R1 (Cheng et al., 2025)	2025	context	trajectory	tool/environment feedback per interacti...	
Mem0 (Chhikara et al., 2025)	2025	ext. memory	–	LLM-driven extraction, consolidation an...	
Debate or Vote (Choi et al., 2025)	2025	context	output	peer arguments in multi-agent debate ro...	○
Language and Experience (Colas et al., 2025)	2025	world model	–	Bayesian inference over executable worl...	○
ARPO (Dong et al., 2025a)	2025	context	trajectory	token entropy spike after tool feedback...	
Active Confusion Expression (Du et al., 2025)	2025	written	output	confusion indicators (e.g. 'tricky'/'co...	○
SoMi-ToM (Fan et al., 2025)	2025	context	–	new first-person multimodal observation...	○
GiGPO (Feng et al., 2025b)	2025	–	trajectory	episode-level group reward plus step-le...	
ReTool (Feng et al., 2025a)	2025	context	trajectory	outcome reward only; no per-step or sta...	
AgentRefine (Fu et al., 2025)	2025	context	trajectory	environment observation triggers correc...	
Survey of self-evolving agents (Gao et al., 2025)	2025	–	–	none	
AccumulatingContext (Geng et al., 2025)	2025	context	–	beliefs drift as context accumulates th...	✓
DIAMONDS (Ghosh et al., 2025)	2025	–	–	values of tracked quantities change ove...	✓
GenEnv (Guo et al., 2025)	2025	world model	trajectory	alpha-Curriculum Reward aligns simulato...	
Martingale Score (He et al., 2025)	2025	context	–	new evidence across iterative reasoning...	✓
Measuring Sycophancy of Language... (Hong et al., 2025)	2025	context	–	sustained user pressure across turns	○
MemoryAgentBench (Hu et al., 2025c)	2025	–	–	incremental multi-turn information inje...	✓
Memory in the Age of AI Agents (Hu et al., 2025d)	2025	ext. memory	–	memory formation, evolution and retriev...	

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 19 | Master table, continued (4/10).

System	Year	State kept by	Credited	Revision trigger	Metric
Agent2World (Hu et al., 2025b)	2025	world model	trajectory	testing-team unit tests and simulation-...	○
Text2World (Hu et al., 2025a)	2025	world model	output	none (one-shot world-model generation,...	○
Bayesian Coherence Coefficient (Imran et al., 2025)	2025	context	–	in-context evidence pieces presented wi...	✓
Search-R1 (Jin et al., 2025b)	2025	context	output	retrieved documents interleaved with re...	
SearchRLStudy (Jin et al., 2025a)	2025	context	trajectory	outcome reward plus format reward; inte...	
Emergent World Beliefs (Kamel et al., 2025)	2025	world model	–	none (probing of pretrained activations)	✓
ACON (Kang et al., 2025b)	2025	written	–	compression guidelines refined in natur...	○
MemoryOS (Kang et al., 2025a)	2025	ext. memory	–	dialogue-chain FIFO from short- to mid-...	○
CoEx (Kim and won Hwang, 2025)	2025	world model	–	subgoal experience assimilated into neu...	
RefAct (Kim et al., 2025)	2025	written	–	each step rewrites a natural-language s...	○
LLMs get lost multi-turn (Laban et al., 2025)	2025	context	–	new user information across turns (mode...	○
Hindsight (Latimer et al., 2025)	2025	ext. memory	–	reflect operation updates evolving beli...	
CWM games (Lehrach et al., 2025)	2025	world model	–	unclear (model synthesised from rules a...	○
DeepAgent (Li et al., 2025e)	2025	ext. memory	interaction	autonomous memory folding compresses in...	
Word to World (Li et al., 2025f)	2025	world model	WM loss	next-state prediction under interaction	○
Turn-PPO (Li et al., 2025c)	2025	context	interaction	turn-level advantage estimation in turn...	
SALT (Li et al., 2025b)	2025	–	interaction	none	
SORL (Li et al., 2025a)	2025	context	interaction	turn-level importance sampling; clippin...	
MemOS (Li et al., 2025g)	2025	ext. memory	–	memory lifecycle scheduling: MemCubes w...	
AgentFlow (Li et al., 2025h)	2025	ext. memory	trajectory	verifier output appended to evolving me...	
RecToM (Li et al., 2025d)	2025	–	–	questions require tracking evolving int...	○
ABELL (Lidayan et al., 2025)	2025	written	state	belief grading: ground-truth grader or...	○
iStar (Liu et al., 2025)	2025	context	interaction	implicit PRM trained by trajectory-leve...	
M3-Agent (Long et al., 2025)	2025	ext. memory	trajectory	streaming visual/auditory input continu...	○
ARPO (Lu et al., 2025)	2025	context	trajectory	none	
Agent Lightning (Luo et al., 2025b)	2025	context	interaction	credit assignment module decomposes tra...	
UltraHorizon (Luo et al., 2025a)	2025	context	–	none	○
R-WoM (Mei et al., 2025)	2025	world model	–	retrieved tutorial knowledge grounds si...	○
MATPO (Mo et al., 2025)	2025	–	trajectory	credit assigned across planner and work...	
MindGames (Moore et al., 2025)	2025	–	–	agent must ask about the interlocutor's...	○
LLaMAR (Nayak et al., 2024)	2025	context	–	none	
WMNav (Nie et al., 2025)	2025	world model	–	discrepancy between world-model plan an...	○
ReasoningBank (Ouyang et al., 2025)	2025	ext. memory	–	self-judged success/failure of each tas...	○
ID-RAG (Platnick et al., 2025)	2025	ext. memory	–	unclear (identity KG queried each decis...	○
Lookbacks belief tracking (Prakash et al., 2025)	2025	context	–	visibility lookback updates observing c...	○
LLM Debate Overconfidence (Prasad and Nguyen, 2025)	2025	context	–	new opponent arguments each round (shou...	✓
ToolIRL (Qian et al., 2025)	2025	context	output	reward on tool-call format and correctn...	
Bayesian teaching (Qiu et al., 2026a)	2025	written	state	distillation of a normative Bayesian mo...	✓
Zep / Graphiti (Rasmussen et al., 2025)	2025	ext. memory	–	temporally-aware KG edge invalidation w...	○
Cognitive Memory survey (Shan et al., 2025)	2025	ext. memory	–	none (survey; covers updating, conflict...	
TTI (Thinking vs Doing) (Shen et al., 2025)	2025	context	trajectory	additional environment observations fro...	
ToMATO (Shinoda et al., 2025)	2025	context	–	information asymmetry between role-play...	✓
WMACT (Shu et al., 2025)	2025	context	output	none (outcome reward)	
R4 (Sohn et al., 2025)	2025	world model	–	continuous 4D database construction fro...	
R1-Searcher (Song et al., 2025)	2025	context	trajectory	none (search results appended to contex...	
ContextFolding (Sun et al., 2025b)	2025	written	memory op	agent branches into a sub-trajectory an...	
ZeroSearch (Sun et al., 2025a)	2025	context	trajectory	none (outcome reward on final answer; s...	
Dynamic Cheatsheet (Suzgun et al., 2025)	2025	ext. memory	memory op	self-curation after each query: model w...	○
TARL (Tan et al., 2025b)	2025	context	interaction	LLM-judge turn-level adjudication	
MemBench (Tan et al., 2025a)	2025	ext. memory	–	none (evaluation only)	○
LLM cooperative game simulation (Tao et al., 2025)	2025	ext. filter	–	Bayesian belief updating plus noisy rep...	○
NextLat belief-state latents (Teoh et al., 2025)	2025	world model	WM loss	auxiliary next-latent prediction loss;...	○

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 20 | Master table, continued (5/10).

System	Year	State kept by	Credited	Revision trigger	Metric
RAGEN / StarPO (Wang et al., 2025f)	2025	context	trajectory	environment reward; reasoning-aware rew...	
VAGEN (Wang et al., 2025d)	2025	written	state	world-modeling reward on turn-level sta...	○
IGPO (Wang et al., 2025a)	2025	context	interaction	turn-level information gain: marginal i...	○
ENACT (Wang et al., 2025e)	2025	world model	–	none (evaluation only)	○
EMPG (Wang et al., 2025c)	2025	context	interaction	step-wise policy entropy modulates grad...	○
Memento 2 stateful reflective mem... (Wang, 2025)	2025	ext. memory	memory op	memory write = policy evaluation, read...	
SPA-RL (Wang et al., 2025b)	2025	–	interaction	none	
StepSearch (Wang et al., 2025g)	2025	context	interaction	per-step information gain and redundanc...	
Fine-Grained Reward Credit (Wei et al., 2025a)	2025	context	interaction	per-turn environment rewards in turn-le...	
WebAgent-R1 (Wei et al., 2025d)	2025	context	trajectory	none (binary task-success reward only)	
GTR-Turbo (Wei et al., 2025c)	2025	–	trajectory	none	
Evo-Memory (Wei et al., 2025b)	2025	ext. memory	–	memory refined after each task in a seq...	○
ReSum (Wu et al., 2025c)	2025	written	trajectory	periodic tool-based summarization when...	
RLVR-World (Wu et al., 2025a)	2025	world model	WM loss	RL reward on decoded transition-predict...	○
CollabLLM (Wu et al., 2025b)	2025	context	interaction	none for belief; Multiturn-aware Reward...	
DEL-ToM (Wu et al., 2025d)	2025	written	state	each DEL belief-update step scored by a...	✓
AgentPRM (Xi et al., 2025)	2025	context	interaction	TD/GAE-estimated promise and progress l...	
Experience-following memory study (Xiong et al., 2025b)	2025	ext. memory	memory op	addition gated by trajectory evaluators...	○
FLAG-TRADER (Xiong et al., 2025a)	2025	context	interaction	policy-gradient RL on trading reward ov...	
A-Mem (Xu et al., 2025e)	2025	ext. memory	output	memory evolution: new notes trigger upd...	
STARE-VLA (Xu et al., 2025a)	2025	–	interaction	none	
PPR (Xu et al., 2025c)	2025	context	trajectory	none	
EnigmaToM (Xu et al., 2025b)	2025	written	–	iterative masking over entity-state sce...	○
Memory-R1 (Yan et al., 2025)	2025	ext. memory	memory op	learned ADD/UPDATE/DELETE/NOOP operatio...	
EvolvTrip (Yan et al., 2025b)	2025	written	–	narrative events append to a perspectiv...	○
ToM-agent counterfactual reflecti... (Yang et al., 2025a)	2025	written	–	counterfactual gap between predicted an...	○
ECON (Yi et al., 2025)	2025	ext. filter	trajectory	TD-loss update of a learned belief netw...	
MemAgent (Yu et al., 2025b)	2025	written	state	chunk-wise overwrite of fixed-size memo...	
Dyna-Think (Yu et al., 2025c)	2025	world model	WM loss	DDT stage 1 trains state prediction / s...	✓
Demystifying agentic RL (Yu et al., 2025d)	2025	context	trajectory	outcome reward on end-to-end tool-use t...	
PersuasiveToM (Yu et al., 2025a)	2025	–	–	questions probe shifts of desires/belie...	○
EvolveCast (Yuan et al., 2025b)	2025	context	–	post-cutoff evidence presented; forecas...	✓
MARSHAL (Yuan et al., 2025a)	2025	–	interaction	turn-level advantage estimator aligned...	
Agentic RL survey (Zhang et al., 2025d)	2025	–	–	n/a	
CriticSearch (Zhang et al., 2025i)	2025	–	interaction	none	
LLMs as discounted Bayesian filte... (Zhang et al., 2025e)	2025	context	–	sequential stochastic observations unde...	✓
ACE (Zhang et al., 2025g)	2025	written	–	reflector extracts lessons from executi...	○
Memory in LLMs: mechanisms/eval (Zhang et al., 2025c)	2025	ext. memory	–	write->read->inhibit/update chain; mode...	○
ReasonRAG (Zhang et al., 2025j)	2025	context	interaction	process-level rewards on query generati...	
Ark / MedArk (Zhang et al., 2025a)	2025	context	trajectory	ask-vs-tell decision via self-reasoning...	○
MindPower (Zhang et al., 2025h)	2025	written	output	Mind-Reward rewards consistency between...	○
Beyond Markovian (Zhang et al., 2025i)	2025	context	trajectory	observed outcomes update a posterior ov...	
PRIME (Zhang et al., 2025k)	2025	ext. memory	–	new user engagements update episodic me...	
Sycophancy under Pressure (Zhang et al., 2025f)	2025	context	output	adversarial user pressure in dialogue;...	○
Overcoming Multi-step Complexity... (Zhang et al., 2025b)	2025	ext. filter	output	stepwise Bayesian updates with small-LM...	○
PrefEval (Zhao et al., 2025)	2025	context	–	user preference stated explicitly or im...	○
DeepResearcher (Zheng et al., 2025c)	2025	context	trajectory	outcome reward on final answer from end...	○
LifelongAgentBench (Zheng et al., 2025b)	2025	ext. memory	–	cross-task experience replay; group sel...	○
MEM1 (Zhou et al., 2025d)	2025	written	state	internal state rewritten each turn cons...	
Memento (Zhou et al., 2025a)	2025	ext. memory	interaction	environment feedback rewrites episodic...	
SWEET-RL (Zhou et al., 2025c)	2025	–	interaction	step-level critic reward using training...	
Social World Models (S3AP) (Zhou et al., 2025b)	2025	written	–	structured state/action/mental-state re...	○

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 21 | Master table, continued (6/10).

System	Year	State kept by	Credited	Revision trigger	Metric
T3 (Zou et al., 2025)	2025	prob. memory	trajectory	truth-anchored potential -log b(s*) sta...	✓
MEDLEY-BENCH (Abtahi et al., 2026)	2026	context	–	social disagreement / analyst summaries...	
Agent-ToM: learning to monitor ag... (Ahmed and Nafisi, 2026)	2026	–	trajectory	deviation from task-consistent expected...	
Can LLM Agents Price Competitivel... (Ahmed et al., 2026)	2026	context	–	market feedback and demand shocks revis...	○
Bayesian Chronicle Agents (BCA) (Akbar et al., 2026)	2026	ext. filter	–	Bayesian update (Beta pseudo-counts, pr...	✓
Why world models: LLM state-track... (Alaswad et al., 2026)	2026	context	–	none	
RLAW: POMDP routing + self-correc... (Ali and Tseng, 2026)	2026	ext. memory	interaction	critique score (logic/safety check)	
Bayesian orchestration (Amin, 2026)	2026	ext. filter	state	Bayes-rule posterior update with LLM-el...	○
Clue possibility matrix (Ansell and Toney-Wails, 2026)	2026	ext. filter	–	symbolic constraint update (YES/NO/MAYB...	✓
CA-BED (Arnould et al., 2026)	2026	prob. memory	–	LLM-estimated likelihoods of user answe...	○
RoVLaP (Azran et al., 2026)	2026	ext. filter	–	VLM next-token predicate probabilities...	
BAG (Baan et al., 2026)	2026	context	–	user clarification in the next turn res...	○
RB-VLA: recursive belief VLA (Bagaria et al., 2026)	2026	world model	state	self-supervised world-model objective (...)	○
PatchWorld (Bai et al., 2026)	2026	world model	WM loss	counterexample-guided repair of executa...	○
LLM + causal model POMDP planning (Basu et al., 2026)	2026	ext. filter	–	POMDP observation update; hybrid confid...	○
OmniToM explicit belief modeling (Bawatneh et al., 2026)	2026	–	–	n/a	✓
MIST (Bensal et al., 2026)	2026	ext. memory	–	user misconceptions written into memory...	✓
ECCBench (Berman and Deng, 2026)	2026	–	–	none	✓
CAGE-2 compound agent design (Bogdanov et al., 2026)	2026	ext. memory	–	deterministic state-tracking layer with...	
Reasoning Theater (Boppa et al., 2026)	2026	context	output	internal activation belief shift during...	○
InKH financial harness (Borjigin et al., 2026)	2026	ext. memory	–	passive knowledge injection from user/m...	
CPS statistical control survey (Cai et al., 2026)	2026	–	–	none	
TRACE (Chang and Chang, 2026)	2026	written	–	append-only revision when world-model c...	
Trivium (Chang, 2026)	2026	world model	WM loss	budgeted probes and continued posterior...	✓
STALE (Chao et al., 2026)	2026	ext. memory	–	implicit conflict	✓
Belief-State Engine (Chattopadhyay and Halder, 2026)	2026	ext. filter	–	Bayesian posterior	✓
The Horizon Gap (survey) (Chen et al., 2026b)	2026	–	–	n/a	
BCO: belief-calibrated scaffold o... (Chen et al., 2026c)	2026	written	state	evaluation of new candidate scaffolds (...)	✓
MM-Lifelong / ReMA (Chen et al., 2026a)	2026	ext. memory	–	dynamic memory management iteratively u...	○
Theory of Agent (ToA) (Cheng et al., 2026)	2026	–	–	none	
Truth stance layer (expressed dou... (Cochinescu, 2026)	2026	prob. memory	–	rule-based revision operator (user corr...	✓
BOND (distilling Bayesian beliefs) (Cui and Lin, 2026)	2026	written	state	teacher posterior	✓
Closing the Feedback Loop (Cui et al., 2026)	2026	ext. memory	–	realised returns interpreted by an LLM...	
STOCKTAKE (Deb and Krishnan, 2026)	2026	–	–	n/a	✓
MDLM text world models (Deshpande, 2026)	2026	world model	WM loss	world model conditions on state anchors...	○
DeltaLogic minimal-edit revision (Dhanda, 2026)	2026	context	output	minimal premise edit	✓
HIPIF (Diao et al., 2026)	2026	written	trajectory	subgoal completion (history folding) +...	
SGCD sibling-guided credit (Ding et al., 2026b)	2026	–	output	none	
APEx (Ding et al., 2026a)	2026	ext. memory	memory op	reward-guided distillation of trajector...	
Belief at Risk (Dixon, 2026b)	2026	ext. filter	–	Bayesian posterior over regimes	✓
POMDP validation framework (Dixon, 2026c)	2026	ext. filter	–	Bayesian	✓
Governance-aware POMDP for AI del... (Dixon, 2026a)	2026	ext. filter	–	Bayesian update on evidence quality	
Mind-Studio (Dong et al., 2026)	2026	world model	–	entropy-selected state-action-next-stat...	✓
LifeSim (Duan et al., 2026)	2026	context	–	user simulator driven by BDI model; ass...	○
KbSD (Feng et al., 2026)	2026	context	output	none	○
MedExAgent (Gao et al., 2026a)	2026	context	output	none (composite outcome reward: accurac...	
Auditing belief-conditioned Werew... (Gao et al., 2026b)	2026	ext. memory	state	logged per-turn belief updates	✓
CubeBench (ang Gao et al., 2025)	2026	–	–	n/a (benchmark)	○
Cyclic subtask graphs (Gharzeddine and Saab, 2026)	2026	written	–	state-analysis-and-routing agent applic...	
FinBench calibration benchmark (Ghosh and Devarakonda, 2026)	2026	–	output	none	
CHAL: council of hierarchical age... (Giovannelli and Kent, 2026)	2026	written	state	gradient-informed revision on thesis st...	
Does Theory of Mind Improvement R... (Gong et al., 2026)	2026	context	–	live user responses in interactive firs...	○

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 22 | Master table, continued (7/10).

System	Year	State kept by	Credited	Revision trigger	Metric
ENIGMA (Gritsaenko, 2026)	2026	prob. memory	–	deterministic rule-based belief crystal. . .	✓
HiDiffTIR (Guo et al., 2026c)	2026	context	interaction	group tool-call success rate	
PivoARL (Guo et al., 2026b)	2026	context	interaction	structured reflection identifies pivota. . .	
CAVE (Guo et al., 2026a)	2026	context	interaction	process reward decomposed into belief u. . .	○
EnterpriseArena (Han et al., 2026)	2026	context	–	none	
EVAF (Han, 2026)	2026	ext. memory	memory op	surprise- and valence-gated consolidati. . .	○
Structured memory over-personaliz. . . (Hannoon et al., 2026)	2026	ext. memory	–	none (inference-time restructuring of m. . .	○
MemOps (Hao et al., 2026)	2026	ext. memory	–	lifecycle ops (remember/forget/update/r. . .	✓
YC-Bench (He et al., 2026b)	2026	written	–	scratchpad persistence across context t. . .	
TRACE turn credit for jailbreaking (He et al., 2026d)	2026	context	interaction	leave-one-turn-out semantic masking	
HGPO (He et al., 2026c)	2026	–	interaction	none	
CoLearn (He et al., 2026a)	2026	prob. memory	–	Bayesian update (soft-evidence BKT; LLM. . .	✓
CAPTURE: preference drift vs memo. . . (Hossain et al., 2026)	2026	world model	state	uncertainty-triggered clarification + c. . .	✓
REDEREF: training-free probabilis. . . (Hosseini et al., 2026)	2026	prob. memory	interaction	judge feedback updates Thompson-samplin. . .	
WMF-AM working memory probe (Hou et al., 2026)	2026	context	–	K sequential operations within one quer. . .	✓
MATTRL test-time multi-agent RL (Hu et al., 2026b)	2026	ext. memory	interaction	turn-level credit for experience pool c. . .	
BRAID (Hu et al., 2026c)	2026	–	interaction	none	
Infinite-Parameter LLM (Hu et al., 2026a)	2026	ext. filter	state	known answer (gold passage relevance la. . .	✓
SDPO (Hübotter et al., 2026)	2026	context	output	textual environment feedback (errors, j. . .	
ToMAgent lookahead ToM (Hwang et al., 2025)	2026	written	state	dialogue lookahead rollouts score (ment. . .	○
MARBO (Hwang et al., 2026)	2026	written	state	cross-entropy against ground-truth role. . .	○
Persuasion Propagation (Jeong et al., 2026)	2026	context	–	none	○
Planted latent variable (Jerpelea, 2026)	2026	context	–	Bayesian posterior over the planted lat. . .	✓
EOG belief propagation (Jha et al., 2026)	2026	ext. filter	–	new evidence labelled by the LLM (cause. . .	○
Hindsight Memory-PRM (Jia et al., 2026)	2026	ext. memory	memory op	hindsight audit trail (retrieval hits,. . .	○
SCRIBE (Jiang and Ferraro, 2026)	2026	context	interaction	skill-prototype rubric reward on subgoa. . .	
HAGE (Jiang et al., 2026b)	2026	ext. memory	memory op	RL on downstream task jointly updates e. . .	
PABU (Jiang et al., 2026c)	2026	written	state	SFT on LLM-synthesised progress and ret. . .	
Jev-Mem (Jiang et al., 2026a)	2026	ext. memory	–	System-One controller types, links and. . .	
Regent Chess replication (Joshi, 2026)	2026	written	–	none (elicited belief scored only)	✓
SuS strategy-aware surprise (Kashirskiy and Makarov, 2026)	2026	context	output	surprise residual (pre-post prediction. . .	
Context Gathering Decision Process (Kausik et al., 2026)	2026	ext. filter	–	n/a	
InfoPO (Kong et al., 2026)	2026	context	interaction	information-gain reward: credit to turn. . .	○
Belief-Based World Model (Kumar et al., 2026)	2026	world model	–	world-model belief	
Hindsight (Latimer et al., 2026)	2026	ext. memory	–	retain/reflect operations; opinions car. . .	○
MINTEval (Lee et al., 2026a)	2026	–	–	none (evaluation of memory under update. . .	✓
Meta-Harness (Lee et al., 2026b)	2026	ext. memory	memory op	outer-loop search over harness code usi. . .	
Belief geometries via SAEs (Levinson, 2026)	2026	context	–	none	○
Text World Models (review) (Li et al., 2026d)	2026	world model	–	n/a	
ECPO (Li et al., 2026e)	2026	–	trajectory	n/a	
SERL selective hindsight distilla. . . (Li et al., 2026c)	2026	context	interaction	per-step environment feedback (error me. . .	
VICT (Li et al., 2026b)	2026	–	interaction	verifier atom proof edges	
InfoDelphi (Li et al., 2026f)	2026	context	–	private evidence surfaced through peer. . .	✓
SE-Search (Li et al., 2026a)	2026	ext. memory	memory op	memory purification at each Think-Searc. . .	
RTPO (Li et al., 2026g)	2026	–	interaction	reverse-turn tree updates align each tu. . .	
ELPO (Liang et al., 2026)	2026	–	interaction	first irrecoverable step located by bin. . .	
Belief Memory (BeliefMem) (Liao et al., 2026a)	2026	ext. memory	–	Noisy-OR update	
TCPO turn-level credit (Liao et al., 2026b)	2026	context	interaction	verifier score-to-credit conversion; hi. . .	
UQ for LLM Agents taxonomy (Lin et al., 2026c)	2026	–	–	n/a	✓
SkillC (Lin et al., 2026a)	2026	context	trajectory	skill-vs-no-skill reward gap	
ReAdapt (Lin et al., 2026b)	2026	written	–	LLM Adapt step after each tool observat. . .	
MADE: belief-driven dual-agent de. . . (Liu et al., 2026e)	2026	ext. memory	interaction	execution failure -> root-cause invalid. . .	

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 23 | Master table, continued (8/10).

System	Year	State kept by	Credited	Revision trigger	Metric
PiCA pivot-based credit (Liu et al., 2026b)	2026	context	trajectory	pivot-step potential-based shaping	
SHADOW (Liu et al., 2026g)	2026	context	interaction	dynamics-aware state grouping and local...	
INTENT (Liu et al., 2026c)	2026	world model	–	inference-time: intention-aware hierarc...	○
SimpleMem (Liu et al., 2026d)	2026	ext. memory	–	online semantic synthesis merges relate...	○
WorldMemArena (Liu et al., 2026a)	2026	ext. memory	–	gold memory updates and distractors ann...	✓
PRISM (Liu et al., 2026h)	2026	written	–	new observations and VLM semantic reaso...	
MMPO (Liu et al., 2026i)	2026	written	state	belief-entropy proxy (own answer entrop...	
REE diagnostic for ICL in games (Liu et al., 2026f)	2026	context	–	none (interaction history in context)	
AutoHarness (Lou et al., 2026)	2026	ext. filter	–	environment feedback (illegal-move erro...	○
HISR (Lu et al., 2026c)	2026	context	interaction	segment-level process reward modulated...	
UI-Copilot (Lu et al., 2026b)	2026	ext. memory	interaction	memory decoupling separates persistent...	
PaW (Lu et al., 2026a)	2026	world model	WM loss	clipped-MAE auxiliary loss on next-obse...	○
From Storage to Experience (memor... (Luo et al., 2026c)	2026	ext. memory	–	reflection	
Graph of States (Luo et al., 2026d)	2026	ext. memory	–	symbolic check (causal graph constraint...	
ReflectiChain world-model supply... (Luo, 2026)	2026	world model	trajectory	latent trajectory rehearsals; reflection...	
ReflectiChain (Luo et al., 2026a)	2026	world model	–	retrospective reflection over the execu...	✓
PUMA user-state modeling (Luo et al., 2026b)	2026	prob. memory	WM loss	new user utterance updates variational...	✓
ProPlay (Ma et al., 2026b)	2026	world model	trajectory	post-episode graph refinement from envi...	
Fine-Mem (Ma et al., 2026a)	2026	ext. memory	memory op	fine-grained feedback alignment on memo...	
LongHorizon-Harness (Ma et al., 2026c)	2026	ext. memory	–	none	
CoMAM (Mao et al., 2026b)	2026	ext. memory	memory op	none	
TreeMem (Mao et al., 2026a)	2026	ext. memory	memory op	none	
Few-shot in-context world represe... (Matsutani et al., 2026)	2026	context	–	none (in-context inference)	✓
Physics of long-horizon planning (Men et al., 2026)	2026	world model	WM loss	simulator ground-truth state transition...	○
EquiMem (Meng et al., 2026)	2026	ext. memory	–	each memory update scored against equil...	○
PCCC / CVT-RL (Meng, 2026)	2026	written	state	verifier-estimated prefix-observable la...	✓
Episodic-Semantic Memory (Milosevic, 2026)	2026	ext. memory	memory op	consolidation of episodic window into s...	○
MT-GRPO Iterative Reward Calibrat... (Modercua et al., 2026)	2026	context	interaction	none (per-turn reward calibrated from r...	
Bayesian Linguistic Forecaster (Murphy, 2026)	2026	written	–	sequential update in tool loop	✓
BeliefShift (Myakala et al., 2026)	2026	–	–	evidence-driven revision	✓
When State Constraints Go Uncheck... (Nakayashiki, 2026)	2026	ext. memory	–	budgeted verification	
DToM-Track: dynamic ToM as tempor... (Nguyen et al., 2026)	2026	context	–	none	✓
Light-Omni (Nie et al., 2026)	2026	ext. memory	–	hierarchical merging consolidates episo...	
EvoHarness-RL learned harness sta... (Ning et al., 2026)	2026	ext. memory	interaction	learned read/update/consolidate policy...	
SSMG-Nav (Niu et al., 2026)	2026	ext. memory	–	new observations consolidated into a pe...	○
SkillIOS (Ouyang et al., 2026)	2026	ext. memory	memory op	trainable curator updates SkillRepo fro...	
MD5 state tracking (Pai and Xian, 2026)	2026	context	–	majority vote over arithmetic	
Business Arena marketplace benchm... (Pan et al., 2026a)	2026	–	interaction	none	
ContextPilot (Pan et al., 2026b)	2026	context	memory op	context and entropy variation identify...	
Bayes-consistent orchestration (Papamarkou et al., 2026)	2026	ext. filter	–	Bayesian posterior update over task-rel...	
Kumiho (Park, 2026)	2026	ext. memory	–	AGM-style revision on graph memory (imm...	○
HiPER: hierarchical RL with expli... (Peng et al., 2026a)	2026	context	trajectory	none (reward-based)	
Explore More Drift Less (Pu et al., 2026)	2026	–	output	n/a	
STAPO trajectory-neglect steps (Qi et al., 2026)	2026	context	trajectory	normalized-entropy outlier steps	
Agents fail to use world models (Qian et al., 2026)	2026	world model	–	agent decides when to invoke external s...	○
MoM / P-Mem (Qin and Lu, 2026)	2026	written	–	LLM typed write-time operations (accept...	✓
SWIRL (Qiu et al., 2026b)	2026	world model	WM loss	alternating GRPO updates of forward wor...	○
MatchTIR (Qu et al., 2026)	2026	context	interaction	bipartite matching of predicted vs grou...	
Debugging code world models (Rahmani, 2026)	2026	world model	WM loss	predicted runtime state after each exec...	✓
Oblivion (Rana et al., 2026)	2026	ext. memory	memory op	read gated by agent uncertainty and buf...	
LLM explainer over Active Inferen... (Raval et al., 2026)	2026	ext. filter	–	variational (Active Inference) postero...	✓
IAPO: influence-aware credit for... (Ren et al., 2026a)	2026	context	interaction	none (influence-dependency graph from r...	

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 24 | Master table, continued (9/10).

System	Year	State kept by	Credited	Revision trigger	Metric
CSSP (Ren et al., 2026b)	2026	ext. filter	state	new telemetry events update a learned s...	○
AI Scientists Reasoning (Ríos-García et al., 2026)	2026	context	–	refutation-driven revision (rarely obse...	✓
BayesBench (Samanta et al., 2026)	2026	context	–	per-turn evidence arrival; compared aga...	✓
PCE (Seo et al., 2026)	2026	written	state	scenario likelihood scoring	
FinalityBench (Sharma, 2026)	2026	context	–	none (benchmark; agent probes/waits)	
SPADER (Shi et al., 2026)	2026	context	interaction	none	
Agent-BRACE (Singh et al., 2026b)	2026	written	state	verbalized certainty	✓
Nous: when belief-based memory he... (Singh, 2026)	2026	prob. memory	state	surprise (information-theoretic) + reli...	✓
MMA-RAG*T (Singh et al., 2026a)	2026	written	–	new observations at each pipeline check...	○
Pinductor: LLM-prior POMDP world... (Six et al., 2026)	2026	world model	state	belief-based likelihood score of candid...	
Parallel WebBench (Sogani et al., 2026)	2026	context	trajectory	GRPO on task outcome; trace diagnostics...	○
D-MEM (Song and Xin, 2026)	2026	ext. memory	–	reward prediction error gating	
Hybrid-WM (Song and Cai, 2026a)	2026	world model	state	consistency gate: disagreement between...	✓
WM-SAR (Song and Cai, 2026b)	2026	world model	trajectory	spectral error amplification	
Semantic Bayesian World Models (Soru, 2026)	2026	ext. filter	state	Bayesian conditioning on observations;...	
CMI (Srivastava, 2026b)	2026	ext. memory	state	counterfactual answer-score intervention	
memory-bench construction (Srivastava, 2026a)	2026	ext. memory	–	hidden ledger of facts defines expected...	✓
External affective state dynamics (Subaharan, 2026)	2026	ext. memory	–	exponential smoothing or momentum updat...	
When Memory Updates but Behavior... (Sun and He, 2026)	2026	ext. memory	–	state dependency repair	
VerMem (Sun et al., 2026b)	2026	ext. memory	memory op	local verifier scores each memory trans...	○
AEWM / EditAct (Sun et al., 2026a)	2026	world model	interaction	teacher or judge (annotation-agent labe...	
HCAPO hindsight critic (Tan et al., 2026)	2026	context	trajectory	post-hoc LLM critic refinement	
MSCE (Tang et al., 2026a)	2026	ext. memory	trajectory	reflection-weighted value backfilling p...	○
ReBel (Tang et al., 2026b)	2026	written	state	belief-consistency reward: each predica...	○
Proactive Service Agents survey (Tang et al., 2026c)	2026	–	–	Bayesian belief update (formal framewor...	
STAGE (Tian et al., 2026)	2026	written	state	new scene evidence triggers recursive u...	✓
Memora (Uddin et al., 2026)	2026	ext. memory	–	user-stated add/update/delete of prefer...	✓
Rank-Bounded Memory (Verbovoy, 2026)	2026	ext. memory	–	storage schema invariants (ownership pa...	✓
TOKI (Wang, 2026b)	2026	ext. memory	–	bitemporal operators (4 heuristics)	
CAPO: critic-guided action-aligne... (Wang et al., 2026a)	2026	context	interaction	none (critic value at action boundary)	
SLEA-RL: step-level experience au... (Wang and Jiang, 2026)	2026	ext. memory	trajectory	observation change triggers step-level...	
G2PO (Wang et al., 2026j)	2026	context	trajectory	globally standardized TD error over sta...	
Self-Guide internal reward co-evo... (Wang et al., 2026g)	2026	written	trajectory	self-generated guidance as internal rew...	
TAMTRL (Wang et al., 2026d)	2026	written	trajectory	teacher likelihood of memory update	
T2PO (Wang et al., 2026b)	2026	context	interaction	token-level entropy/confidence plateau	
CrEST (Wang et al., 2026k)	2026	context	interaction	verified turn reward with self-teacher...	
MINDGAMES live arena (Wang et al., 2026c)	2026	–	–	n/a	
APPO (Wang et al., 2026h)	2026	–	output	none	
LENS (Wang et al., 2026f)	2026	prob. memory	–	LLM relevance-oracle verdicts update a...	○
Mind2Dialogue (Wang et al., 2026l)	2026	written	output	simulator updates a shared evolving use...	○
Dark Room (Wang, 2026a)	2026	written	state	rule-verified next-observation predicti...	○
Conv-FinRe (Wang et al., 2026i)	2026	–	–	n/a (benchmark; investor risk preferenc...	○
GAVEL (Wang et al., 2026e)	2026	ext. filter	state	known answer (RSN prior over object roo...	✓
Agentic Reasoning survey (Wei et al., 2026)	2026	–	–	n/a	
AC1 self-model loop replication (Wilson, 2026)	2026	ext. filter	–	Bayesian accumulator over own interacti...	✓
Bayesian-Agent (Wu et al., 2026b)	2026	ext. filter	–	verified trajectories	
BayesEvolve: explicit belief for... (Wu et al., 2026d)	2026	ext. filter	state	posterior refit after each evaluated ca...	✓
HumanLM state alignment (Wu et al., 2026a)	2026	written	state	RL aligns generated natural-language la...	○
RL recipe long-horizon tool agents (Wu et al., 2026c)	2026	context	trajectory	staged vs dense reward shaping on Trave...	
REFLEX (Jev) (Wu and Lim, 2026)	2026	context	–	confidence threshold escalates bounded...	
SkillIRL (Xia et al., 2026)	2026	ext. memory	trajectory	experience distilled into hierarchical...	
WebWorld (Xiao et al., 2026b)	2026	world model	WM loss	trained on 1M+ open-web interactions; u...	○

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 25 | Master table, continued (10/10).

System	Year	State kept by	Credited	Revision trigger	Metric
AeroBelief (Xiao et al., 2026a)	2026	prob. memory	–	rule-based accumulation of VLM match co...	
TIPS (Xie et al., 2026)	2026	context	interaction	teacher answer-likelihood gain	
LLM Agents for Forecasting (surve... (Xu et al., 2026d)	2026	–	–	n/a	
Contextual Belief Management (Bel... (Xu et al., 2026a)	2026	written	state	symbolic verification	✓
NeSyFS: neuro-symbolic fast-slow... (Xu and Fekri, 2026)	2026	ext. memory	interaction	repeated fast-action failure triggers s...	
GPM: governed persistent memory (Xu, 2026)	2026	ext. memory	–	symbolic check (source binding, retract...	
MemGym (Xu et al., 2026c)	2026	ext. memory	–	none	
EnvACE world rehearsal (Xu et al., 2026e)	2026	world model	trajectory	task-success RL reward jointly optimize...	
META (Xu et al., 2026b)	2026	ext. memory	–	trade outcome + LLM reflection; prune/c...	
Representation signatures in trad... (Xue, 2026)	2026	context	output	structured risk feedback	
MemStrata temporal validity (Yadav, 2026)	2026	ext. memory	–	contradiction of a stored (subject, rela...	✓
Memory-R2 / LoGo-GRPO (Yan et al., 2026a)	2026	ext. memory	state	local rerollouts from shared intermedia...	
DenoiseFlow (Yan et al., 2026b)	2026	context	trajectory	semantic entropy over samples	
Belief Engine stance dynamics (Yang et al., 2026c)	2026	prob. memory	state	log-odds update with evidence uptake an...	✓
GroupMemBench multi-party memory (Yang et al., 2026b)	2026	ext. memory	–	none	
Graph-based agent memory survey (Yang et al., 2026a)	2026	ext. memory	–	memory evolution stage (update/consolid...	
EvoTool (Yang et al., 2026e)	2026	context	interaction	none	
world models as intermediary (Yang, 2026)	2026	world model	WM loss	none (position paper)	
EnterpriseBench (Yang et al., 2026d)	2026	context	–	none (benchmark)	
Werewolf accusation belief-shift (Yang et al., 2026f)	2026	written	–	LLM in-context update after accusation...	✓
Forecast-Dojo (Ye et al., 2026)	2026	written	trajectory	LM rewrites JSON belief notebook (proba...	✓
AgeMem (Yu et al., 2026b)	2026	ext. memory	memory op	agent-initiated store/retrieve/update/s...	○
LOCARD structured belief state (Yu and Knottenbelt, 2026)	2026	written	–	strategic core flips a boolean SOP-mile...	
InfAgent (Yu et al., 2026a)	2026	written	–	none	
SAVeR: self-audited verified reas... (Yuan et al., 2026a)	2026	context	state	adversarial audit locating logical/evid...	
RPMS (Yuan et al., 2026b)	2026	written	–	lightweight belief state gates memory a...	○
Critic Experience Bank (Zeng et al., 2026)	2026	ext. memory	–	hindsight LLM majority vote conditioned...	✓
Credit Assignment in RL for LLMs... (Zhang, 2026)	2026	–	trajectory	n/a	
MICA: intertemporal credit for em... (Zhang et al., 2026d)	2026	ext. memory	interaction	none (potential over structured user st...	
Q-Evolve (Zhang et al., 2026k)	2026	context	trajectory	advantage from in-distribution IQL crit...	
Internalizing the Future (Zhang et al., 2026h)	2026	written	trajectory	foresight-conditioned RL calibrating pl...	
RetailBench long-horizon retail P... (Zhang et al., 2026c)	2026	context	–	none	
WorldEvolver (Zhang et al., 2026g)	2026	world model	state	prediction-observation mismatch	
Theory of Space (Zhang et al., 2026f)	2026	written	–	new partial observations during active...	✓
MemSkill (Zhang et al., 2026b)	2026	ext. memory	memory op	controller selects extract/consolidate/...	○
OASES (Zhang et al., 2026a)	2026	context	state	none	
HiMem (Zhang et al., 2026e)	2026	ext. memory	–	conflict detected between retrieved not...	○
StarWM (Zhang et al., 2026i)	2026	world model	WM loss	offline next-observation prediction fro...	○
CORE (+PERSIST) (Zhang et al., 2026j)	2026	prob. memory	state	known answer (gold slot values: state-f...	✓
AEM adaptive entropy modulation (Zhao et al., 2026a)	2026	context	trajectory	response-level surprisal x advantage	
EvoSCM causal belief revision (Zhao et al., 2026b)	2026	written	state	prediction-observation discrepancy dist...	✓
BranPO (Zhao et al., 2026c)	2026	–	interaction	none	
PGPO (Zheng et al., 2026c)	2026	context	interaction	potential difference between consecutiv...	
FOREAGENT predict-before-execute (Zheng et al., 2026a)	2026	world model	–	predict-then-verify: prediction replace...	○
Code2World (Zheng et al., 2026b)	2026	world model	WM loss	render-aware RL rewards rendered next-s...	○
Thinking-RFT for ToM (Zhong et al., 2026a)	2026	–	output	verifiable-reward RL on shortcut-free T...	○
LSE-MTP (Zhong et al., 2026b)	2026	context	WM loss	multi-token prediction anchored to grou...	○
ToolVerse (Zhou et al., 2026)	2026	context	interaction	turn-aware relative advantage from outc...	
Simple mechanism interfaces (Zhu et al., 2026)	2026	context	–	none	
Qwen-AgentWorld (Zuo et al., 2026)	2026	world model	WM loss	next-state prediction reasoning trained...	✓

Metric: ✓ belief-level metric reported, ○ proxy only.

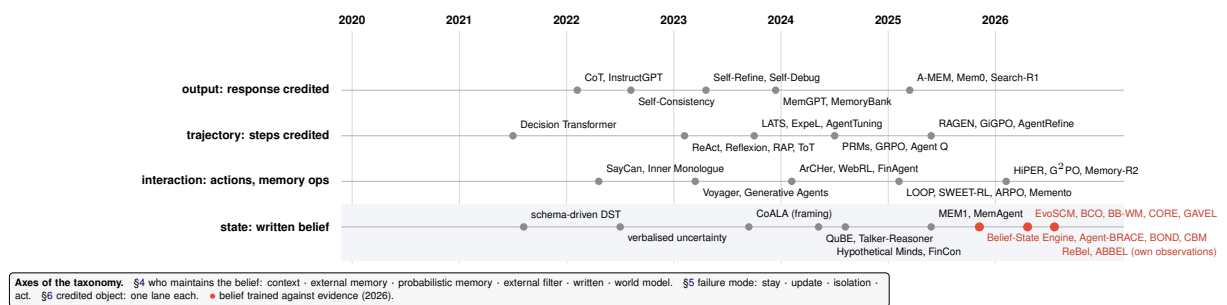


Fig. 6 | Where credit has gone, 2020–2026. Lanes are credited objects; Table 15 lists every screened system. Written beliefs exist from dialogue state tracking onward, but until 2026 they are credited by task outcome or not trained at all; the red points are the first to give the belief its own learning signal.